



The therapeutic potential of Ziziphi Spinosae Semen and Polygalae Radix in insomnia management: Insights from gut microbiota and serum metabolomics techniques

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ABSTRACT

Ethnopharmacological relevance: Ziziphi Spinosae Semen and Polygalae Radix (ZSS-PR) constitute a traditional Chinese herbal combination with notable applications in clinical and experimental settings due to their evident sedative and calming effects. Aligned with traditional Chinese medicine principles, Ziziphi Spinosae Semen supports cardiovascular health, nourishes the liver, and induces mental tranquillity. Simultaneously, Polygalae Radix elicits calming effects, fosters clear thinking, and reinstates proper coordination between the heart and kidneys. ZSS-PR is commonly employed as a therapeutic intervention for various insomnia types, demonstrating distinct clinical efficacy. Our previous study findings provide evidence that ZSS-PR administration significantly reduces sleep onset latency, increases overall sleep duration, and improves abnormal neurotransmitter levels in a murine insomnia model.

Aim of study: This investigation aimed to scrutinize the intrinsic regulatory mechanism of ZSS-PR in managing insomnia using gut microbiota and serum metabolomics techniques.

Materials and methods: Mice were given DL-4-Chlorophenylalanine to induce insomnia and then treated with ZSS-PR. The open-field test assessed the animals' spontaneous activity. Concentrations of neurotransmitters, endocrine hormones, and cytokines in the duodenum were measured using enzyme linked immunosorbent assay, and brain histopathology was evaluated with H&E staining. The impact of ZSS-PR on the metabolic profile was examined by liquid chromatography coupled to high resolution mass spectrometry, and 16S rDNA sequencing was used to study the influence of ZSS-PR on the gut microbiota. Additionally, the content of short-chain fatty acids (SCFAs) was analyzed by GC-MS. Finally, correlation analysis investigated relationships between biochemical markers, metabolites, SCFAs, and gut microbiota.

Results: ZSS-PR treatment significantly increased movement time and distance in mice with insomnia and improved pathological impairments in the cerebral cortex and hippocampus. It also restored abnormal levels of biochemical markers in the gut of insomnia-afflicted mice, including 5-hydroxytryptamine, dopamine, gastrin, melatonin, tumour necrosis factor- α , and interleukin-1 β . Metabolomics findings showed that ZSS-PR had a significant restorative effect on 15 endogenous metabolites in mice with insomnia. Furthermore, ZSS-PR primarily influenced five metabolic pathways, such as phenylalanine, tyrosine, and tryptophan biosynthesis, glutamine, and glutamate metabolism. Additionally, gut microbiota analysis revealed notable alterations in both diversity and microbial composition after ZSS-PR treatment. These changes were primarily attributed to the relative abundances of microbiota, including Firmicutes, Bacteroidota, Fusobacteriota, *Muribaculaceae unclassified*, and *Ligilactobacillus*. The results of SCFAs analysis demonstrated that ZSS-PR effectively restored abnormal levels of acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, and valeric acid in insomniac mice. Subsequent correlation analysis revealed that microbiota show obvious correlations with both biochemical markers and metabolites.

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Conclusions: The results provide compelling evidence that ZSS-PR effectively mitigates abnormal activity, reduces cerebral pathological changes, and restores abnormal levels of neurotransmitters, endocrine hormones, and cytokines in mice with insomnia. The underlying mechanism is intricately linked to the modulation of gut microbiota and endogenous metabolic pathways.

1. Introduction

Insomnia, characterized by difficulty falling asleep, shortened sleep time, and shallow sleep (Chen and Tang, 2021), often accompanies reduced memory and attention (Xu et al., 2019). Approximately one-third of the global population faces sleep problems, with 4–6% meeting clinical diagnostic criteria for insomnia (Chung et al., 2015). Prolonged insomnia can induce endocrine disorders, resulting in cognitive impairments, cardiovascular diseases, and other endocrine and immune disorders (Budhiraja et al., 2011; Javaheri and Redline, 2017), seriously impacting physical and mental health. In clinical practice, sedative-hypnotic drugs like benzodiazepine-receptor agonists, melatonin and variants, antidepressants, antipsychotics, and antihistamines are commonly prescribed for insomnia treatment (Cunnington et al., 2013). However, these drugs have limitations, including single-target action, drug resistance, addiction, withdrawal reactions, and limited efficacy against different insomnia types (Zhang and Xia, 2011). Traditional Chinese medicine (TCM) is gaining attention as a complementary and alternative therapy due to its advantages of minimal side effects, low risk of developing dependency, and ability to target multiple areas. TCM views insomnia as related to imbalances in Yin and Yang, disharmony between nutrient Qi and defensive Qi, and organ function imbalances. The holistic treatment approach involves techniques such as syndrome differentiation and treatment (Singh and Zhao, 2017). For instance, addressing restlessness stemming from stomach disharmony involves methods to harmonize the stomach and promote mental calmness, yielding favorable clinical outcomes (Zhang and Feng, 2014).

Ziziphi Spinosae Semen (ZSS) refers to the dried mature seeds of *Ziziphus jujuba* Mill., which belongs to the Rhamnaceae family. ZSS is recognized for its beneficial effects on the heart and liver. It nourishes these organs, calms the mind, and promotes body fluid production while reducing excessive sweating. ZSS is notably effective in addressing restless sleep caused by a deficiency of heart and liver blood, making it a frequent prescription for various insomnia types. Polygalae Radix (PR) designates the dried roots of either *Polygala tenuifolia* Willd. or *Polygala sibirica* L., which belongs to the Polygalaceae family. PR possesses diverse therapeutic properties, including mind-calming, cognitive function improvement, phlegm dissipation, expectoration promotion, and swelling reduction. Commonly used for insomnia, dreaminess, forgetfulness, and palpitations arising from heart and kidney imbalances, PR is a well-established sedative herb in Traditional Chinese Medicine (TCM). In clinical practice, PR is often combined with ZSS to treat insomnia, anxiety, and related conditions, with notable formulations such as Guipi Tang and Qifuyin. Some pharmacological data from animal studies, have demonstrated that ZSS possesses various pharmacological effects. These effects encompass sedative and hypnotic properties (Jiang et al., 2007), antidepressant activity (Li et al., 2022b), anxiolytic effects (Chen et al., 2023), and neuroprotection. The main active substances in ZSS, including flavonoids (e.g., spinosin), saponins (such as jujuboside A and jujuboside B), and alkaloids (e.g., magnolflorine), phenylpropanoids (e.g., coumestrol), and alkanolic acids are believed to underlie its sedative and hypnotic effects (Li et al., 2022c; Ren et al., 2023). Similarly, PR has demonstrated various

pharmacological effects, such as memory improvement (Ren et al., 2022a), sedative and hypnotic properties (Sun et al., 2023), antidepressant activity (Chen et al., 2021), anxiolytic effects (Wen, 2006), and neuroprotection (Wang et al., 2023). The primary active ingredients in PR, such as polygala saponins, are deemed responsible for their sedative and calming effects (Cao et al., 2016). When combined with ZSS, PR exhibits a synergistic effect, further enhancing its sedative and hypnotic properties (Luo et al., 2020; Ren et al., 2022b). However, the components mentioned above in both ZSS and PR have limited content and bioavailability, presenting challenges in crossing the blood-brain barrier. Consequently, the specific mechanism of action and potential synergistic effects of Ziziphi Spinosae Semen and Polygalae Radix (ZSS-PR) in treating insomnia remain undetermined.

Accumulating evidence suggests a communication link between the brain and gut, termed the brain-gut axis. The gut microbiota can activate this axis, influencing neural, endocrine, and immune pathways that regulate physiological functions such as sleep, cognition, and disease progression (Li et al., 2020; Zhu et al., 2017). On one hand, the gut microbiota contributes to synthesizing neurotransmitters like 5-hydroxytryptamine (5-HT) and dopamine (DA). These neurotransmitters can directly or indirectly impact the central nervous system through the blood-brain barrier or vagus nerve, thereby regulating sleep. On the other hand, the interaction between the gut microbiota and the gut leads to the release of various cytokines and chemokines by gut secretory cells. These substances act on specific immune cells and the nervous system, thereby regulating sleep (Sun et al., 2016; Zhang et al., 2014). Conversely, insomnia can alter the structure and function of the gut microbiota through the brain-gut axis. Additionally, it has been observed that short-chain fatty acids (SCFAs) produced by gut microbiota serve as signalling molecules. These metabolites directly influence the central nervous system, gastrointestinal tract, and other peripheral nervous systems, thereby impacting the gastrointestinal and nervous systems through peripheral circulation. Metabolomics, a highly effective approach for systematically analyzing changes in endogenous metabolites due to external stimuli (Ćuperlović-Culf, 2012), aligns with the research methods of TCM, emphasizing the study of holistic and multi-target mechanisms. Additionally, gut microbiota dysbiosis can worsen metabolic disorders, possibly arising from short-term sleep deprivation (Benedict et al., 2016). Previous research indicates that extracts from ZSS and PR significantly impact endogenous metabolic regulation (Duan et al., 2022; Heng et al., 2022; Luo et al., 2020). However, the precise mechanism by which ZSS-PR induces sedation and tranquillity via the mentioned neurological and endocrine indicators, along with its regulatory processes, remains unclear in the current academic literature.

This study aims to investigate the mechanism of ZSS-PR intervention in treating insomnia by utilizing metabolomics and gut microbiota analysis. Specifically, we examined the impact of ZSS-PR on the metabolic profile of animals with insomnia by UHPLC-HRMS metabolite profiling. Additionally, we assessed the influence of ZSS-PR on the composition and structure of gut microbiota in these animals through 16S rDNA high-throughput sequencing technology. The effect of ZSS-PR on the production of SCFAs in insomnia-affected animals was analyzed using gas chromatography-mass spectrometry (GC-MS). By integrating

these findings, this study aims to unravel the intricate interrelationships among biochemical factors, metabolic profiles, gut microbiota, and SCFAs. Ultimately, our goal is to provide insights into the regulatory mechanism of the brain-gut axis and the therapeutic potential of ZSS-PR in managing insomnia.

2. Materials and methods

2.1. Materials

Ziziphi Spinosae Semen, the dried mature seeds of *Ziziphus jujuba* Mill., Lot # 210301, and Polygalae Radix, the root of *Polygala tenuifolia* Willd., Lot # 210427, were procured from Beijing Tongrentang Health Pharmaceutical Co., Ltd (Beijing, China) and authenticated by Dr. Tan Jinyan from Shanxi University of Chinese Medicine, following the standards of the Chinese Pharmacopoeia 2020 edition ([Chinese Pharmacopoeia Commission, 2020](#)). Voucher specimens were deposited in the Herbarium of Shanxi University of Chinese Medicine (voucher number: 202103021224294-220310; 202103021224294-220226). All decoction pieces were within the validity period and had a two-year shelf life. Tween-80, Lot # 150130, was acquired from Beijing Scianen Pharmaceutical Co., Ltd (Beijing, China). Methyl tert-butyl ether, HPLC grade, Lot #C12166741, was obtained from Macklin Inc. (Macklin, Shanghai, China). Hematoxylin and Eosin staining, Lot # 20220425, were purchased from Phygene Life Sciences Co., Ltd (Fuzhou, China). Deionized water (DI water), 18.2 MΩ, was prepared using a Milli-Q purification system. Mouse 5-HT (5-Hydroxytryptamine) ELISA KIT, Lot # AD221010, DA (dopamine) ELISA KIT, Lot # AD221011, Gas (gastrin) ELISA KIT, Lot # AD221017, MT (melatonin) ELISA KIT, Lot # AD221029, TNF-α (tumour necrosis factor-α) ELISA KIT, Lot # AD221030, and IL-1β (interleukin-1β) ELISA KIT, Lot # AD221009, were sourced from Andy Gene Biotechnology Co., Ltd (Beijing, China). Pusion® Hot start flex 2X Master Mix, Lot #M0536L, was purchased from Shanghai Yitao Biological Instrument Co. Ltd (Shanghai, China). DL2000 DNA Marker, Lot # 3427A, was obtained from Takara. Gene color, Lot # GBY-1, was acquired from Beijing Jinboyi Biotechnology Co., Ltd (Beijing, China). Qubit™ dsDNA HS Assay Kit (500 assays), Lot #Q32854, was obtained from Invitrogen Life Technologies (CA, USA). Biowest Agarose G-10, Lot # 111860, was sourced from BIOWEST (Spain). TAE buffer, 50x, Lot #B548101-500, was purchased from Sangon Biotech Co., Ltd (Shanghai, China). NovaSeq 6000 SP Reagent Kit, 500 cycles, Lot # 20028402, was acquired from Illumina (San Diego, CA, USA). Short-chain fatty acids standards (GC grade): Acetic acid, Lot #C12700504; Propionic acid, Lot #C14458542; Isobutyric acid, Lot #C13706285; Butyric acid, Lot #C14253081; Isovaleric acid, Lot #C14414245; and Valeric acid, Lot #C14026528, were purchased from Macklin Biochemical Technology Co., Ltd (Shanghai, China). 2-Ethylbutyric acid, Lot #H2231563, was obtained from Aladdin Biotechnology Co., Ltd (Shanghai, China).

2.2. Extraction and chemical composition determination of ZSS-PR

Following established scholarly literature, the ZSS-PR herbal combination was prepared by accurately measuring the raw materials in a 2:1 ratio, with 50 g of ZSS and 25 g of PR, respectively ([Ren et al., 2022b](#)). The herbal materials were mixed and soaked in 10 times the amount of water for 30 min. Subsequently, the mixture underwent boiling and extraction for 40 min. Afterwards, the mixture was filtered, and the remaining residue was boiled again with 8 times the amount of water for 30 min. The resulting mixture was then filtered and combined with the initial filtrate. The combined filtrates were concentrated to achieve a final concentration of 1 g/mL (i.e., 0.66g of ZSS and 0.33g of PR). A 10 mL ZSS-PR decoction (1 g/mL) was freeze-dried and transferred to a 25 mL volumetric flask, where 50% methanol was added to dissolve it and filled up to the mark. One millilitre of the solution was mixed with 3 mL of 50% methanol. The resulting solution was filtered

through a 0.22 μm microporous filter, and the filtrate was collected for qualitative analysis of the chemical components. The remaining ZSS-PR decoction was divided into three portions and diluted to the desired concentrations (0.25 g/mL, 0.5 g/mL, 1 g/mL) according to therapeutic doses (low, medium, and high). It was then divided into seven portions according to daily usage and stored at −20 °C. Before use, the portions were thawed. Qualitative analysis of the chemical components present in ZSS-PR was conducted using UHPLC coupled to a Quadrupole-Time of Flight mass spectrometer (AB SCIEX, USA). Detailed instrument parameters, chromatographic, and mass spectrometric conditions were as follows:

The separation utilized an ACQUITY UPLC® HSS T3 C18 column (2.1 × 100 mm, 1.8 μm) at a flow rate of 0.3 mL/min, an injection volume of 5 μL, and a column temperature of 40 °C. The mobile phase consisted of a 0.1% formic acid aqueous solution (A) and acetonitrile (B), following a gradient elution program: 0–8 min, 5%–17% B; 8–10 min, 17% B; 10–11 min, 17%–18% B; 11–12 min, 18%–20% B; 12–17 min, 20%–23% B; 17–22 min, 23%–33% B; 22–30 min, 33%–60% B; 30–32 min, 60%–95% B; 32–36 min, 95% B; 36–37 min, 95%–5% B, 37–40 min, 5% B.

ESI mass spectra were acquired under the following conditions in both positive and negative ionization modes: the spray voltage for ESI+ was set to 5.5 kV, and for ESI-, it was set to −4.5 kV. The ion source temperature was maintained at 500 °C. Nitrogen gas served as the auxiliary gas, with the pressure levels for the air curtain gas (CUR), atomization gas (GS1), and auxiliary gas (GS2) set at 30, 55, and 60 psi, respectively. The collision energy (CE) was adjusted to 35 eV for ESI+ and −35 eV for ESI-. The cluster voltage (DP) was set to 80 V for ESI+ and −80 V for ESI-. Employing the information-dependent scanning (IDA) mode, the ten highest peaks with a response value exceeding 100 cps were chosen for the secondary mass spectrometry scan. The scanning range of m/z 50–1500 was utilized, and dynamic background deduction (DBS) was activated to mitigate interference. The acquired data were analyzed and processed using PeakView (v2.2, SCIEX). To identify the chemical components present in ZSS-PR, we compared accurate molecular mass, retention time (tR), and fragment ions provided by PeakView with the compound information in the reference literature.

2.3. Animal ethics statement

Healthy specific pathogen-free (SPF)-grade KM mice (male, 18–20 g) were procured from Sibeifu Biotechnology Co., Ltd, Beijing, China (SCXK (Jing) 2019-0010) and housed under controlled conditions at a temperature of 25 ± 1 °C, a humidity of 60 ± 2%, and a light/dark cycle of 12 h/12 h. The mice underwent a 7-day acclimation period with unrestricted access to standard laboratory pellets and water. The study protocol underwent a thorough review, and all animal experiments strictly adhered to the Guidelines for the Care and Use of Laboratory Animals and the Chinese Legislation on Laboratory Animals. Every effort was made to ensure the well-being of the mice, minimize their suffering, and reduce the number of animals used. Ethical approval for the animal experiments conducted in this research was obtained from the ethical committee of Shanxi University of Chinese Medicine (Grant No. 2021DW228).

2.4. Animal grouping and treatment

After one week of adaptation to the feeding conditions, mice were randomly assigned to six groups, each consisting of six mice. The groups included the control group (referred to as Control), the insomnia model group (referred to as Model), the positive group (referred to as Positive, receiving diazepam at a dosage of 0.83 g/kg, Lot # 211006, from Shandong Xinyi Pharmaceutical Co., Ltd., Dezhou, China), the low-dose ZSS-PR treatment group (referred to as DL, with a dosage of 2.5 g/kg), the medium-dose ZSS-PR treatment group (referred to as DM, with a dosage of 5.0 g/kg), and the high-dose ZSS-PR treatment group (referred

to as DH, with a dosage of 10.0 g/kg). ZSS-PR extract and diazepam suspensions were separately prepared using distilled water. Different ZSS-PR doses were calculated based on the quantity of ZSS-PR raw materials. Dosages were determined in accordance with the clinical practices of Chinese medicine practitioners and literature references (Ren et al., 2022b). Adult individuals' body weight was assumed to be 60 kg, and the dose conversion coefficient between human and mouse administration was set at 10. The human equivalent dose (HED) for the low dose was calculated as 2.5 g/kg, representing 0.5 times the clinical dosage. Similarly, the HED for the medium dose was determined as 5 g/kg, corresponding to the clinical dosage. Likewise, the HED for the high dose was calculated as 10 g/kg, equivalent to twice the clinical dosage.

Starting on the eighth day, intraperitoneal injections of DL-4-Chlorophenylalanine (PCPA, 500 mg/kg, Lot #C13205137, Macklin, China) suspension were administered to all groups, except for the control group, following the cited literature (Hua et al., 2021). The suspension was prepared using 0.1% (w/v) Tween-80 normal saline. This injection was administered once a day for three consecutive days. In contrast, the control group received an injection of normal saline into their abdominal cavity. Subsequently, starting on the eighth day, all groups were orally administered the corresponding medicine once a day for seven consecutive days, with a dose of 0.1 mL/10 g. The treatment groups received ZSS-PR or diazepam through oral administration, while the control and model groups were administered the same volume of distilled water. The mice's daily activities, mental states, fur, and body weight were carefully observed and recorded throughout the experiment. The experimental procedure is depicted in Fig. 1.

2.5. Assessment of autonomic activity of animals

On Day 14, the open-field test (OFT) was conducted to assess the animals' spontaneous activity and anxiety levels, following a previously described protocol (Abdollahnejad et al., 2016). The OFT box was divided into nine equal-sized squares, and mice from each group were individually placed in the central area. The experiment occurred in a well-ventilated, quiet, and dimly lit environment. The OFT-100 video analysis system (Chengdu Taimeng Software Co., Ltd., Chengdu, China) recorded the Rest Time, Move Time, and Movement Distance of the mice within the central area of the open field for 5 min. After each trial, any feces left in the OFT box were promptly removed, and 75% ethanol was used for disinfection to prevent interference in subsequent experiments.

2.6. Collection and processing of experimental animal samples

Upon completion of the experiment, mice were intraperitoneally anaesthetized with a 2% sodium pentobarbital solution (40 mg/kg) prepared with normal saline. Blood samples were then collected from the orbit of the mice, and serum was isolated for metabolomics analysis. The brains were removed and fixed in a 4% paraformaldehyde solution for subsequent histopathological examination after decapitation. The duodenum was harvested for ELISA assay. Colon contents were collected in sterile centrifuge tubes for gut microbiota analysis, and cecal contents were used for the analysis of SCFAs. All samples, except for the brains, were preserved at -80 °C.

2.7. Histological observation of brain tissues

The brain tissues underwent fixation in paraformaldehyde, dehydration in ethanol, embedding in paraffin, and cross-sectioning at 4 or 5 μm thickness. Subsequently, they were stained with Hematoxylin and Eosin (H&E) and observed under an optical microscope (Axioscope 5, ZEISS, Germany) at a magnification 200x.

2.8. Determination of the expression of neurotransmitters, endocrine hormones and cytokines in the duodenum

Duodenal tissues were homogenized in a 10% (w/v) phosphate-buffered saline solution (pH 7.4). The mixture was then centrifuged at a speed of 3000 rpm for 10 min, and the resulting supernatant was collected. ELISA kits were employed, following the manufacturer's instructions, to quantify the expression of neurotransmitters (5-HT, DA), endocrine hormones (Gas, MT), and cytokines (TNF-α, IL-1β) in the duodenum.

2.9. Metabolomics analysis of serum

A total of 100 μL serum was collected and combined with three times the volume of pre-cooled methanol (MS grade, Thermo Fisher Scientific, Lot # 216678) for protein precipitation. The mixture was vortexed for 1 min and then centrifuged at 14,000 rpm for 15 min at 4 °C. The resulting supernatant was collected and concentrated using a stream of nitrogen until dry. Subsequently, 200 μL methanol was added to redissolve the sample, vortexed for 3 min, and centrifuged again at 14,000 rpm for 15 min at 4 °C. Finally, 2 μL of the supernatant was analyzed by UHPLC

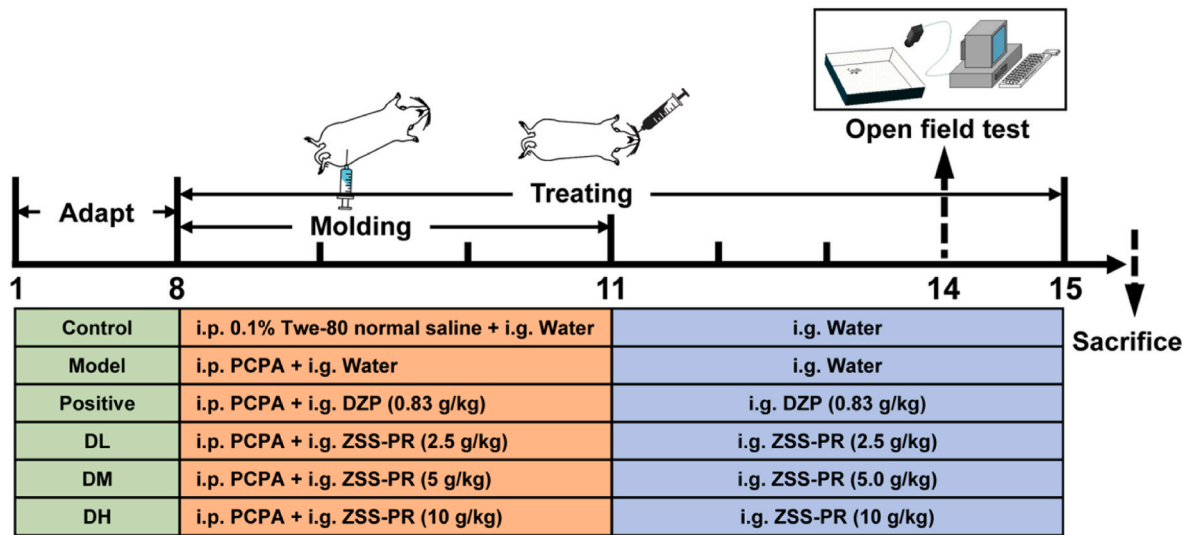


Fig. 1. Schematic Overview of the Study Design (Mice Experiment). I.p. indicates intraperitoneal injection, while i.g. denotes intragastric administration. (Control) Control group; (Model) Model group; (Positive) Positive group; (DL) Low-dose ZSS-PR group; (DM) Medium-dose ZSS-PR group; (DH) High-dose ZSS-PR group.

coupled to a Q Exactive Hybrid Quadrupole-Orbitrap mass spectrometer (Thermo Scientific). For instrument stability, 2 μL of each sample was mixed to create pooled quality control (QC) samples. Before formal analysis, six consecutive QC samples were injected to equilibrate the system. Additionally, one injection was performed after every six samples to monitor the analytical system stability continuously.

The separation was carried out on an ACQUITY UPLC BEH C18 column (2.1 $\times$ 100 mm, 1.7 μm) at a flow rate of 0.3 mL/min, with an injection volume of 5 μL and a column temperature of 40 $^{\circ}\text{C}$. The mobile phase consisted of acetonitrile (MS grade, Thermo Fisher Scientific, Lot # 215625) (A) mixed with a 0.1% formic acid (HPLC grade, Aladdin, Lot #H1913009) aqueous solution (B), following a gradient elution program as follows: 0.0–0.5 min, 5% A; 0.5–1.5 min, 5%–15% A; 1.5–4.5 min, 15%–30% A; 4.5–6.0 min, 30%–60% A; 6.0–9.0 min, 60%–70% A; 9.0–12.0 min, 70%–100% A; 12.0–13.0 min, 100% A; 13–13.5 min, 100%–5% A; 13.5–16.0 min, 5% A. The electrospray ionization (ESI) mass spectra in positive and negative ionization modes were obtained under the following conditions: in positive ion mode, set the spray voltage to 3.2 kV, the sheath gas flow rate to 40 arb, the auxiliary gas flow rate to 5 arb, and the auxiliary gas heating temperature to 350 $^{\circ}\text{C}$. In negative ion mode, set the spray voltage to 2.5 kV, the sheath gas flow rate to 38 arb, the auxiliary gas flow rate to 10 arb, and the auxiliary gas heating temperature to 300 $^{\circ}\text{C}$. Adjust the ion transfer tube temperature to 320 $^{\circ}\text{C}$, and the lens voltage (S-Lens RF Level) to 50 V. For full scan/data-dependent secondary scan (full MS/dd-MS2), use a scanning range of m/z 100–1,000, with a primary mass resolution of 70,000 FWHM and a secondary resolution of 17,500 FWHM. Finally, use a collision energy of 30 eV.

To compile data lists for retention time (RT), mass-to-charge ratio (m/z), and normalized peak area in both positive and negative ion modes, the raw data were imported into Compound Discoverer v. 3.3 (Thermo Fisher Scientific, Waltham, MA, USA). This software facilitated chromatographic peak alignment, matching, and area normalization, with the area normalization method set as total peak area normalization. The results were expressed as peak area percentages. Subsequently, principal component analysis (PCA) and orthogonal partial least squares-discriminant analysis (OPLS-DA) were conducted using Simca-P v. 14.1 (Umetrics Inc., Umea, Sweden) to identify crucial metabolites. Key metabolites were determined based on criteria such as a variable importance plot (VIP) value exceeding one and a significance level (p) below 0.05 (compared with the Model). Mass spectrum information of the critical metabolites was aligned with possible chemical formula information using One-MAP (<http://www.5omics.com/>). The annotation of metabolites was accomplished by comparing their MS data (accurate mass, MS/MS fragments) with reference data in the Human Metabolome Database (HMDB; <http://www.hmdb.ca/>). Pathway enrichment analysis was carried out using MetaboAnalyst 5.0 (<http://www.metabolanalyst.ca/>) and the KEGG database (<http://www.genome.jp/kegg/>).

2.10. Preparation of total DNA and high throughput sequencing analysis

The analysis of animal gut microbiota was conducted using 16S rDNA high-throughput sequencing as follows: total microbial DNA was extracted from colonic contents using the CTAB method (Huang et al., 2018), eluted with 50 μL buffer, and stored at -80°C . Subsequently, the V3–V4 hypervariable region of the bacterial 16S rDNA gene was amplified through thermocycling PCR, utilizing primers 341F (5'-CCTACGGGNGGCWGCAG-3') and 805R (5'-GACTACHVGGGTATCTAATCC-3'). PCR consisted of an initial 30 s denaturation step at 98 $^{\circ}\text{C}$, followed by 32 cycles of denaturation at 98 $^{\circ}\text{C}$ for 10 s, annealing at 54 $^{\circ}\text{C}$ for 30 s, and extension at 72 $^{\circ}\text{C}$ for 45 s, with a final extension step at 72 $^{\circ}\text{C}$ for 10 min. The resulting PCR products were extracted on 2% agarose gels and purified using AMPure XT magnetic beads (Beckman Coulter Genomics, Danvers, MA, USA). Quantification was performed using Qubit (Invitrogen, USA). DNA PCR-Free sample preparation kits were used to generate sequencing libraries and add index codes

according to the manufacturer's instructions. Library quality was assessed using an Agilent 2100 bioanalyzer (Agilent Technologies, Inc., Santa Clara, CA, USA) and a library quantification kit suitable for Illumina (Kapa Biosciences, Woburn, MA, USA). Finally, PE250 sequencing was conducted on an Illumina NovaSeq 6000 sequencer (Illumina, California, USA) with the NovaSeq 6000 SP Reagent Kit (500 cycles, Lot # 20028402, Illumina).

Samples were assigned unique barcodes for paired-end reads, and barcode and primer sequences were clipped. The paired-end reads were merged using the FLASH software and then filtered using fqtrim (v 0.94) criteria to retain only high-quality and useable reads. Further filtering was performed using Vsearch software (v 2.3.4) to remove any chimeric sequences. Clean data were organized into an ASV_table after denoising by DADA2. Subsequently, each feature sequence was annotated through BLAST alignment to the SILVA database (<http://www.arb-silva.de/>), and the ASV_table_taxonomy was generated. The taxa levels were normalized to relative abundance: summing the total counts of each annotated taxon in each sample and then calculating the percentage of the counts for each taxon to the total counts. ASV abundance, taxonomy profile, diversity profile, and metadata were tested in R v 4.1.2, and the Kruskal-Wallis test, followed by Dunn's post-hoc test, was performed for multiple comparisons. To identify significantly different taxa, all characteristic taxa were tested using the Kruskal-Wallis and Wilcoxon test, and the final marker taxa were obtained by combining linear discriminant analysis effect size (LEfSe). Marker taxa were filtered by $P < 0.05$ (corrected by the Benjamini-Hochberg method) and LDA > 4 .

2.11. Determination of SCFAs content in cecal contents

100 mg of cecal contents were mixed with 500 μL ultrapure water and vortexed for 5 min. After centrifugation at 13,000 rpm for 5 min at 4 $^{\circ}\text{C}$, 200 μL of supernatant was collected and mixed with 5 μL of internal standard solution (10 μL 2-ethylbutyric acid dissolved in 1 mL water), 0.1 g of NaCl, and 20 μL of 50% H_2SO_4 . The mixture was vortexed for 1 min, and 200 μL methyl tert-butyl ether (MTBE) was added. After shaking for 1 min, the supernatant was centrifuged at 13,000 rpm for 5 min at 4 $^{\circ}\text{C}$, and the extraction was repeated. The supernatants were pooled together for GC-MS analysis.

Precise amounts of acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, and valeric acid standards were withdrawn using pipettes and transferred into 10 mL volumetric flasks, and dissolved in 10 mL of MTBE. A mixed working standard solution of SCFAs standard (200 mM/L) was prepared by making the volume up to the mark with MTBE. Precise 25 μL of 2-ethylbutyric acid standard was pipetted to a 10 mL volumetric flask and brought to volume with MTBE to create an internal standard mother liquor solution with a 20 mM/L concentration. Finally, 5 mL of the mixed standard master liquor was taken and mixed with 1.67 mL of the internal standard master liquor to obtain the mixed standard solution containing the internal standard. The mixed standard solution of medium concentration was injected six times to analyze precision experiments. Equal amounts of cecal contents were taken and thoroughly mixed to obtain QC samples. Six aliquots of QC samples (100 mg each) were weighed and analyzed using the above method for repeatability experiments. The same QC sample was treated with the same method and tested at 0h, 1h, 6h, 12h, 24h, and 48h time points for stability experiments. The instrument and chromatographic conditions were as follows.

The separation was carried out on an Agilent DB-WAX capillary column (30 m $\times$ 0.250 mm $\times$ 0.25 μm) with helium (99.999%, 1 mL/min) as the carrier gas, and the column temperature initiated at 70 $^{\circ}\text{C}$ and subsequently ramped up to 180 $^{\circ}\text{C}$ at a rate of 10 $^{\circ}\text{C}/\text{min}$. Further, the temperature was elevated to 200 $^{\circ}\text{C}$ at a rate of 5 $^{\circ}\text{C}/\text{min}$. A split ratio of 10:1 was applied. The detector temperature, injection port, transmission line, and ion source temperatures were set at 260 $^{\circ}\text{C}$, 175 $^{\circ}\text{C}$, 280 $^{\circ}\text{C}$, and 230 $^{\circ}\text{C}$, respectively. The injection volume was 1 μL , and the analysis time for each sample was 15 min. A solvent delay time

of 3 min was implemented. The ionization voltage stood at 70 eV, and mass data were collected in full scan mode (m/z 40–400).

For qualitative identification of target SCFAs, the Masshunter software (version 10.0.707.0) was employed to compare the sample Mass spectrum obtained through automatic analysis with the standard mass spectrogram database. 2-ethylbutyric acid served as an internal standard with an assigned abundance value of 1. Subsequently, relative abundance values of other SCFAs in each sample and the internal standard were computed. A standard curve was employed to determine the concentration of SCFAs in the sample for quantitative analysis. The content calculation formula used for this purpose was: SCFAs content of the sample ($\mu\text{g}/\text{mg}$) = $(C \times V)/M$, where C denoted the measured concentration, V represented the volume, and M stood for the weight of the sample.

2.12. Integrative analysis

To explore the mechanism of ZSS-PR in addressing insomnia, we performed correlation analysis on biochemical factors (neurotransmitters, endocrine hormones, and cytokines), metabolites, and microbial taxa at the phylum and genus taxonomic level (including differential microbiota and marker microbiota between the Model and DM). Spearman's rank correlation coefficient was employed for this investigation, and the False Discovery Rate (FDR) correction for multiple comparisons was performed using the Benjamini and Hochberg methods. Redundancy analysis (RDA) was performed to explore the correlation between the SCFAs and gut microbiota, and a Permutational multivariate analysis of variance (PERMANOVA) was used to assess the statistical significance.

2.13. Statistical analysis

Statistical tests of the data were conducted using GraphPad Prism software, version 10.0 (Graphpad Software Inc., San Diego, CA, USA). Normal distribution was verified by the Shapiro-Wilk test. One-way analysis of variance (One-way ANOVA) determined significant differences among multiple groups, and the Dunnett multiple-comparison test was performed for the post-hoc test. Kruskal-Wallis test was applied for data with non-normal distribution, and Dunn's test was performed for the post-hoc test. The significance criterion for differences was set at $p < 0.05$, while $p < 0.01$ indicated highly significant differences.

3. Results

3.1. Qualitative analysis of chemical components in ZSS-PR

The total ion chromatogram (Supplementary Fig. 1) and mass spectrum information of ZSS-PR were extracted and compared with references, leading to the annotation of 152 chemical constituents (Sumner et al., 2007). These constituents encompass 52 flavonoids (e.g., spinosin and vicianin II), 39 oligosaccharide esters (e.g., sibiricose A and 3, 6'-disinapoyl sucrose), 26 triterpene saponins (e.g., jujubosides and polygalasaponins), 15 alkaloids (e.g., magnolflorine and caaverine), 10 xanthenes (e.g., polygalaxanthenes and sibiricaxanthenes), 7 organic acids (e.g., ferulic acid and ceanothic acid), two phenylpropanoids (glomeratose B and E), and one terpene (pseudolaroside B) (Supplementary Table 1). ZSS contributed 84 species (Liu et al., 2021; Shen, 2020; Zhang et al., 2016; Zhang, 2019; Zhu et al., 2014), and PR contributed 68 species (Gao et al., 2021; Li et al., 2022a; Lin et al., 2021). The annotated flavonoids and alkaloids originated from ZSS, whereas oligosaccharide esters and xanthenes were specific components unique to PR. Subsequently, the chromatographic peaks of these components were labelled in the total ion chromatogram (Supplementary Fig. 2).

3.2. Effect of ZSS-PR on general biological signs and autonomic activity in insomniac mice

The OFT results revealed that, compared to the control group, model mice spent less time in the center of the open field and experienced a reduced number of crossings through the open field center lattice (Fig. 2A, 2B). Also, this study observed reduced total distance in the model mice, compared to the control group (Fig. 2C). Following treatment with diazepam or diverse ZSS-PR doses, various degrees of alterations in autonomic activity indices within the open field were observed. Compared to the model group, diverse ZSS-PR groups exhibited noteworthy increases in the indices mentioned above. These findings suggest that PCPA-induced insomniac mice displayed reduced autonomic activity and anxiety-like behavior. Nevertheless, oral diazepam or diverse ZSS-PR doses mitigated autonomic activity abnormalities, restoring them to normal levels.

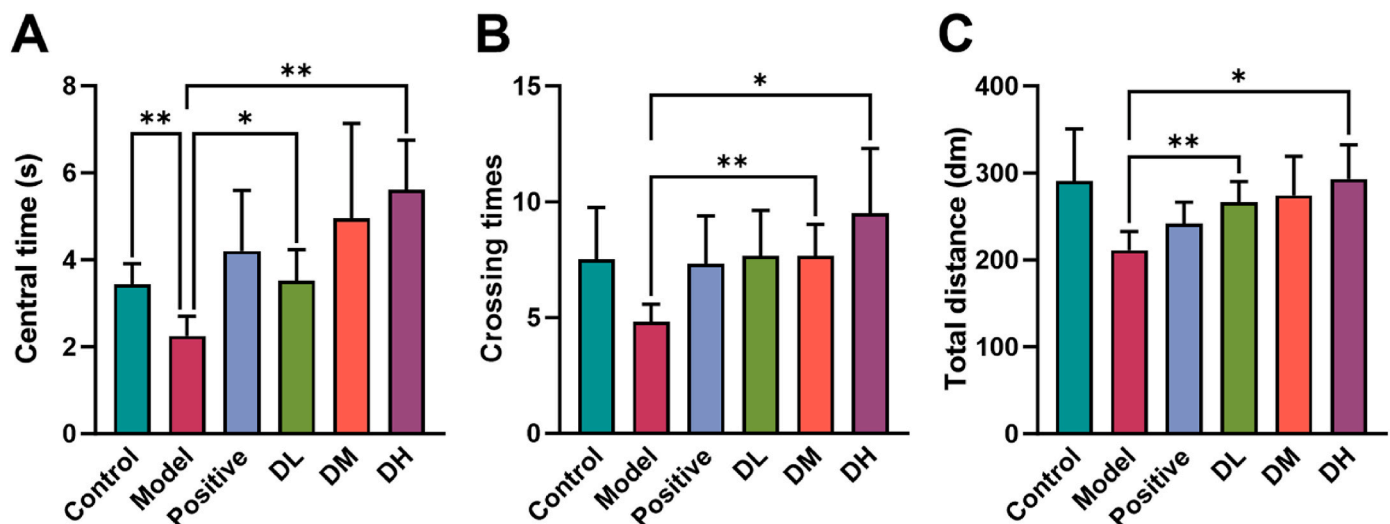


Fig. 2. Evaluation Indices in the Open-Field Test. (A) Central time of mice in each group in the open field. (B) Crossing times of mice in each group in the open field. (C) Total distance of mice in each group in the open field. Values are expressed as mean $\pm$ SD. Normal distribution was confirmed by the Shapiro-Wilk test. Brown-Forsythe and Welch ANOVA, followed by Dunnett's T3 multiple-comparison test, were conducted for statistical analysis. * $p < 0.05$, ** $p < 0.01$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.3. Histological analysis of brain tissue

The hippocampus, located in the temporal lobe, is associated with insomnia, according to a comprehensive assessment of insomnia and brain structure (Stolicyn et al., 2024). It consists of CA1, CA2, CA3, and DG regions, collectively known as the hippocampus. Pathological changes in the cerebral cortex, hippocampal CA1, CA3, and DG regions were observed using a microscope, and the results are presented in Fig. 3. The hippocampal CA1 region displayed structural integrity, with 2–3 layers of neatly aligned and densely packed neurons. The cytoplasm appeared abundant and lightly stained, while the nuclei were well-defined. Similarly, neural cells in the CA3 region exhibited regular shapes, forming numerous abundant and closely aligned layers. The DG region of the hippocampus showcased a clear structure, with pyramidal cells arranged in a regular and orderly manner. In contrast, compared to the control group, cortical neurons exhibited irregular shapes, contractions, and dark staining in the model group. Furthermore, the boundaries between the nucleus and cytoplasm were unclear. The number of neurons in the CA1 region reduced, exhibiting disordered arrangement and diverse cell morphologies. Additionally, some neurons had shrunk. Neural cells in the CA3 region exhibited irregular nuclear bodies, interstitial edema, and indistinct boundaries between the nucleus and cytoplasm. Conversely, cells in the DG region displayed a regular and abundant arrangement. However, some neurons appeared shrunken with unclear boundaries between the nucleus and cytoplasm. Following

treatment with diazepam or different doses of ZSS-PR, the aforementioned pathological features in the mentioned regions exhibited varying degrees of change. Particularly in DH mice, the number of cortical neurons increased, neurons aligned uniformly, and cell morphologies were evident and intact. In the CA3 region, cells were uniformly and homogeneously arranged, displaying complete morphologies and distinct boundaries between the nucleus and cytoplasm. Additionally, their abundance was notable. The DG region comprised many neurons with a regular and orderly arrangement, allowing clear visualization of cell morphology and complete cellular structure. In the DL and DM groups, cells in the CA1 region were regular and tightly aligned, presenting well-defined morphologies and formed structures. The findings revealed mild pathological alterations in the cortex and hippocampus of insomnia-stricken mice. However, treatment with ZSS-PR demonstrated a capacity to mitigate the pathological damage within these specific regions of the insomniac mice’s brains.

3.4. Determination of the expression of neurotransmitters, endocrine hormones and cytokines in the duodenum

Abnormalities in neurotransmitter levels (5-HT, DA), endocrine hormones (MT, Gas), and cytokines (TNF- α , IL-1 β) are commonly associated with the development of insomnia. Fig. 4 illustrates that the model group, compared to the control group, exhibited varying degrees of decreased levels for 5-HT, Gas, MT, TNF- α , and IL-1 β . Conversely,

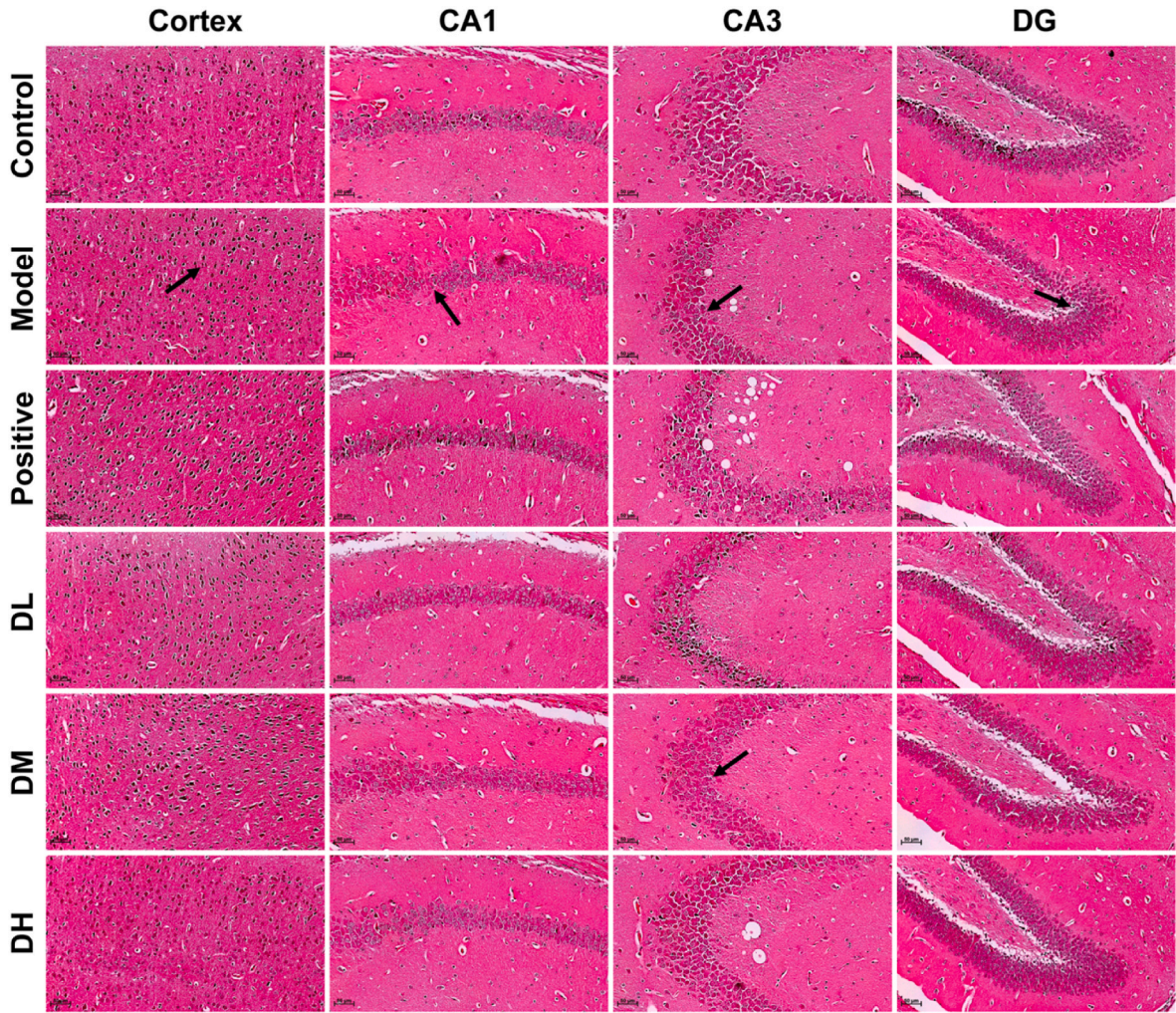


Fig. 3. Pathological Observation of Cerebral Cortex and Hippocampal Regions (CA1, CA3, DG) of the Mice in Each Group (200 × Magnification). The black arrows indicate neuronal cell ruffling.

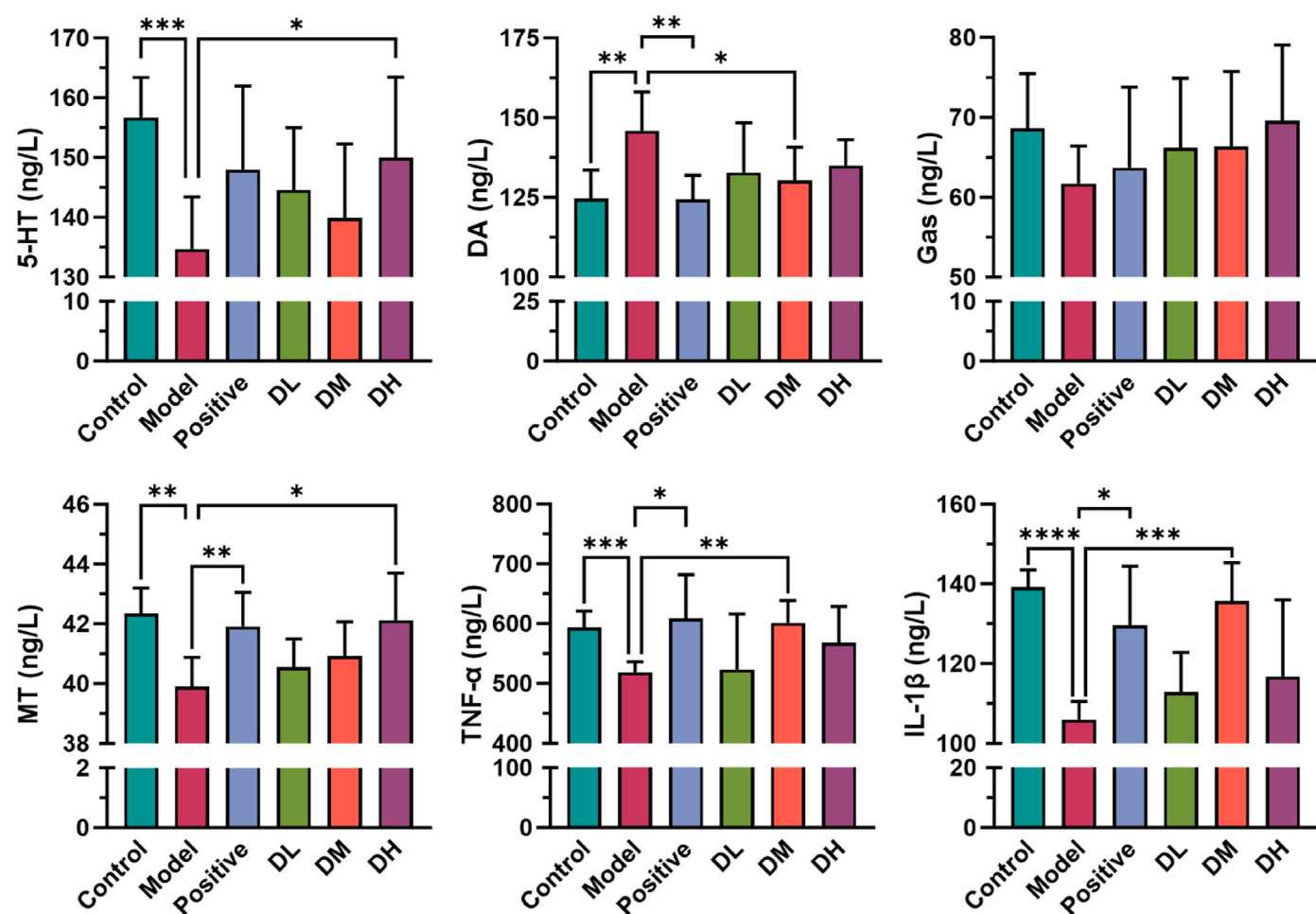


Fig. 4. Effects of ZSS-PR on the Expression of Neurotransmitters (5-HT, DA), Endocrine Hormones (Gas, MT), and Cytokines (TNF- α , IL-1 β) in the Duodenum of Insomniac Mice. Values are expressed as mean $\pm$ SD. Normal distribution was verified by the Shapiro-Wilk test. Brown-Forsythe and Welch ANOVA were performed for statistical analysis, and differences between groups were calculated using an unpaired *t*-test with Welch correction. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

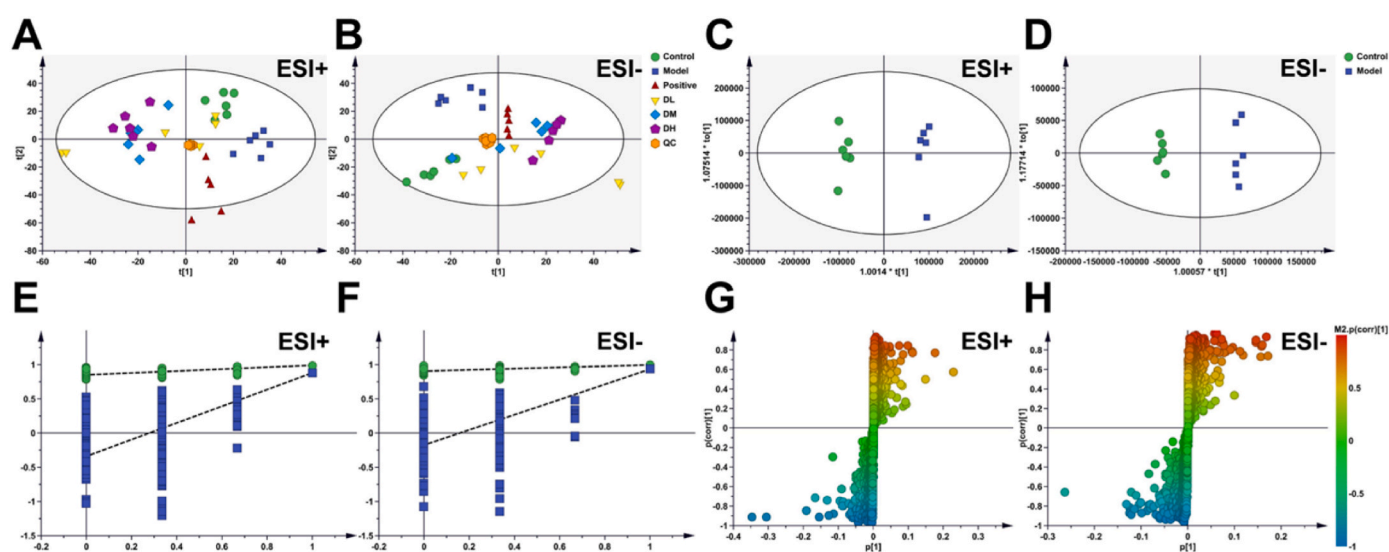


Fig. 5. Metabolomics Analysis of Serum in Both Positive and Negative Ion Modes. ESI + represents the positive ion mode, and ESI- represents the negative ion mode. (A–B) Scatter plot of the PCA scores for each group's samples. (C–D) Scatter plot of OPLS-DA scores comparing Control and Model. (E–F) Permutation displacement test comparing Control and Model (200 times). (G–H) The s-Plot of the Control and Model comparison. The *t*[1] and *t*[2] represent the scores of the first and second principal components, respectively, while the scatter represents the grouping.

there was a significant increase in the level of DA. Treatment with diazepam or ZSS-PR at different doses resulted in different recovery levels in the expression of these indices. Notably, ZSS-PR at different doses had varying regulatory effects on these indices; DM showed notable effects in DA, TNF- α , and IL-1 β levels, while DH led to a significant increase in 5-HT and MT. The results indicate that mice suffering from insomnia exhibit abnormal levels of 5-HT, MT, TNF- α , IL-1 β , and DA in their gut. The administration of different dosages of ZSS-PR treatment effectively alleviates these irregularities in insomnia-afflicted mice.

3.5. Serum metabolome analysis

3.5.1. Metabolic profile analysis

Serum metabolomics studies were conducted, and the total ion chromatogram (TIC) of the QC samples in both positive and negative ion modes is depicted in [Supplementary Fig. 3](#). The serum metabolomic were analyzed using PCA, and the results are presented in [Fig. 5A and B](#). The consistent clustering of the QC samples in both positive and negative ion modes demonstrates the stability and reliability of the analytical method. The Control, Model, Positive, and various doses of ZSS-PR groups were distinctly separated. The samples formed unique clusters within specific ranges under different ion modes within each group. In the $t[2]$ direction, the scatter points of the different ZSS-PR dosage groups were positioned between the control and model groups, suggesting that ZSS-PR effectively corrected metabolic imbalances in PCPA-induced insomniac mice.

To investigate changes in endogenous metabolites in the serum of PCPA-induced insomniac mice and evaluate the impact of ZSS-PR on these metabolites, we conducted OPLS-DA analysis on the model and control groups ([Fig. 5C and D](#)). The model demonstrated excellent performance in the positive ion mode, with $R^2Y = 0.99$ and $Q^2 = 0.88$. Similarly, in the negative ion mode, the model parameters were $R^2Y = 0.995$ and $Q^2 = 0.934$. These values, approaching 1, indicate a high explanatory rate and excellent predictive ability of the model. In the Permutation test (200 times) conducted in both positive and negative ion modes, the left R^2 and Q^2 points consistently demonstrated lower values than the original values on the right. Furthermore, the Q^2 regression line intersected below the origin, affirming the model's validity and absence of overfitting ([Fig. 5E and F](#)). [Fig. 5G and H](#) depict the potential critical metabolites in the PCPA-induced insomnia model, represented by scatter points positioned in the upper left and lower right corners of the s-plot.

3.5.2. Analysis of the key metabolites

The Kruskal-Wallis test was conducted to compare Model Vs. Control and treatment group Vs. Model, respectively. Based on the s-plot, the VIP-values ($VIP > 1$) and Kruskal-Wallis test ($p < 0.05$) were used as criteria to identify significant metabolites. A total of 15 metabolites were identified by comparing them with databases such as mzCloud and HMDB, with 8 in positive ion mode and 7 in negative ion mode ([Supplementary Table 2](#)). To further understand the changes in critical metabolites in insomniac mice and the impact of ZSS-PR on them, the normalized peak area data of the critical metabolites were imported into

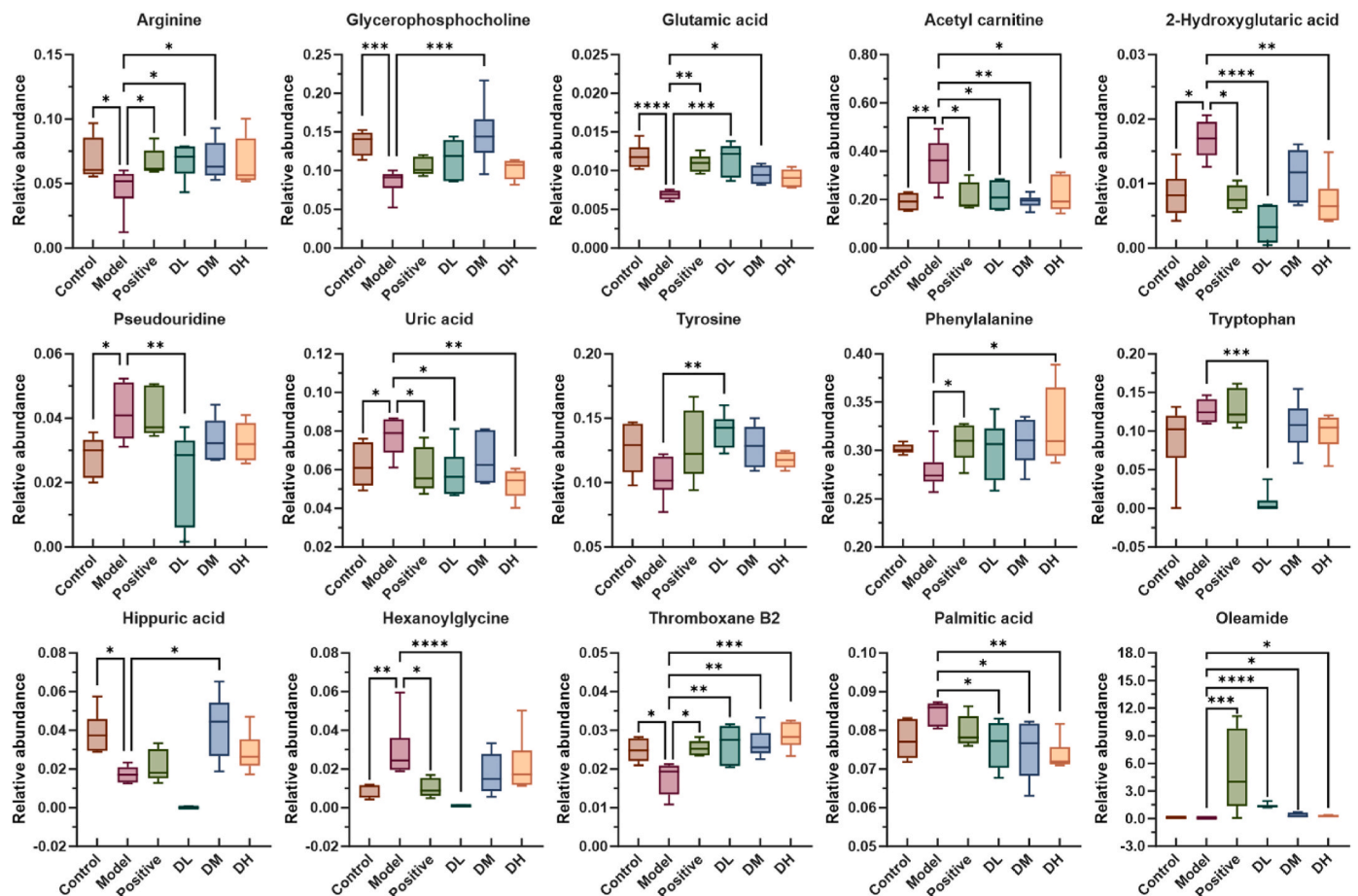


Fig. 6. Relative abundance of potential biomarkers identified in the serum of mice in each group. Relative abundance refers to the normalized peak area. Values within graphs are presented as median values indicating the 25th and 75th percentiles, along with minimum and maximum values (represented as a boxplot), $n = 6$. The significance difference was evaluated by the Kruskal-Wallis test followed by uncorrected Dunn's test, and each comparison stands alone. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

GraphPad Prism v10 (Graphpad Software Inc., San Diego, CA, USA) to generate box plots, as shown in Fig. 6.

Compared to the control group, the serum levels of arginine, glycero-phosphocholine, glutamic acid, tyrosine, phenylalanine, hippuric acid, thromboxane B2, and oleamide in the Model mice were reduced. On the other hand, the levels of acetyl carnitine, 2-hydroxyglutaric acid, pseudouridine, uric acid, tryptophan, hexanoylglycine, and palmitic acid were increased. After treatment with diazepam or various doses of ZSS-PR, the levels of metabolites above exhibited varying degrees of improvement.

3.5.3. Analysis of the metabolic pathways

To investigate the role of ZSS-PR in regulating metabolic pathways, the relevant metabolites were imported into the MetaboAnalyst 5.0 online analysis platform (<https://www.metaboanalyst.ca/>) for enrichment analysis. The resulting data were used to draw a bubble plot, with the pathway Impact factor (*Impact*) on the x-axis and the logarithm of the P-value ($-\log_{10}(P)$) on the y-axis, as shown in Fig. 7A. Additionally, the involvement of key metabolites in critical pathways was elucidated by constructing a metabolite network diagram using the KEGG database (<http://www.genome.jp/kegg/>), as depicted in Fig. 7B.

As illustrated in Figs. 7A and 22 metabolic pathways were enriched in the serum of mice with PCPA-induced insomnia. These pathways were screened using an *Impact* threshold of ≥ 0.01 . The identified pathways include phenylalanine metabolism, phenylalanine, tyrosine, and tryptophan biosynthesis, arginine biosynthesis, arginine and proline metabolism, glutamine and glutamate metabolism, glutathione metabolism, alanine, aspartate, and glutamate metabolism, glycerophospholipid metabolism, tyrosine metabolism, and fatty acid biosynthesis. It suggests that insomnia may be associated with these multiple pathways and the potential role of ZSS-PR in regulating abnormal metabolism in insomniac mice by influencing key metabolites and the aforementioned metabolic pathways. Fig. 7B shows that the chosen biomarkers are highly involved in the pathway network linked to glutamine and glutamate metabolism, arginine biosynthesis, and phenylalanine, tyrosine, and tryptophan metabolism.

3.6. Analysis of the gut microbiota

16S rDNA sequencing was utilized to evaluate each animal group's taxa diversity, composition, and abundance of gut microbiota in each animal group. We identified potential signature microbes, and the results reveal that greater sequencing depth leads to a practically even intestinal microbiota rarefaction curve (Supplementary Fig. 4A), indicating that we have obtained adequate data and met the sequencing criteria. The results genuinely encompass a large portion of microbial information from the samples. Supplementary Fig. 4B presents the outcomes, showcasing each group's cluster analysis of amplicon sequence variants (ASV) types and quantities.

3.6.1. Analysis of the alpha and beta diversity of the gut microbiota

The Chao 1 index directly mirrors the diversity and abundance of microbiota in a sample, with a higher index denoting a more extraordinary microbiota richness. The Shannon and Simpson indices represent the magnitude of gut microbiota diversity, and a higher index signifies increased diversity. Fig. 8A–D illustrates a significant increase ($p < 0.01$) in the model group's Chao1, Shannon, Simpson index of Diversity and Inverse Simpson indices compared to the control group. Upon administering various doses of ZSS-PR, Chao1 and Shannon indices decreased significantly in the DL and DM groups, indicating a trend towards the control group. Compared to the control group, these results signify a substantial rise in the number, richness, diversity, and relative abundance of gut microbiota in mice with insomnia. Conversely, ZSS-PR administration led to reduced number of taxa, richness, diversity, and relative abundance, akin to levels observed in the control group.

The beta diversity analysis outcomes are depicted in Fig. 8E–F. On PCA2, Control samples were closely grouped, while Model samples displayed an upward deviation, making them distinguishable from Control samples. However, treatment groups did not show a significant separation trend compared to Model samples (Fig. 8E). In Fig. 8F, along the PCoA1 axis, the control group exhibited significant dissimilarity compared to the model group, displaying high levels of differentiation, while treatment groups were located between the Control and Model. These results signify a significant alteration in the composition of the microbial community in insomniac mice. After treatment with varying

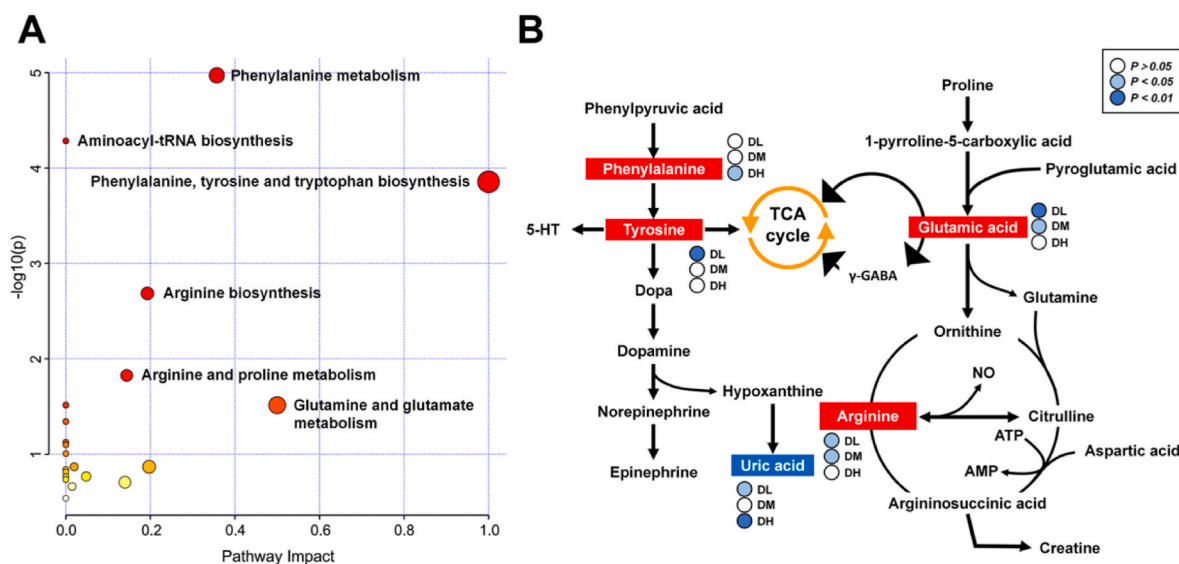


Fig. 7. The Metabolic Pathways and Regulatory Networks Associated with Potential Serum Markers Affected by ZSS-PR in Treating Insomnia. (A) Metabolic pathways involved in potential serum markers are represented by the intensity of color, with the color becoming redder as the $-\log_{10}(P)$ value increases, indicating a higher level of significance for the metabolite. Furthermore, a more significant *Impact* value signifies a greater impact on the pathway. (B) Metabolites labelled in red indicate a significant upregulation caused by ZSS-PR, while those labelled in blue indicate a significant downregulation. The circles symbolize the strength of significance, with dark blue denoting high significance ($p < 0.01$), light blue indicating significance ($p < 0.05$), and white representing no significance ($p > 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

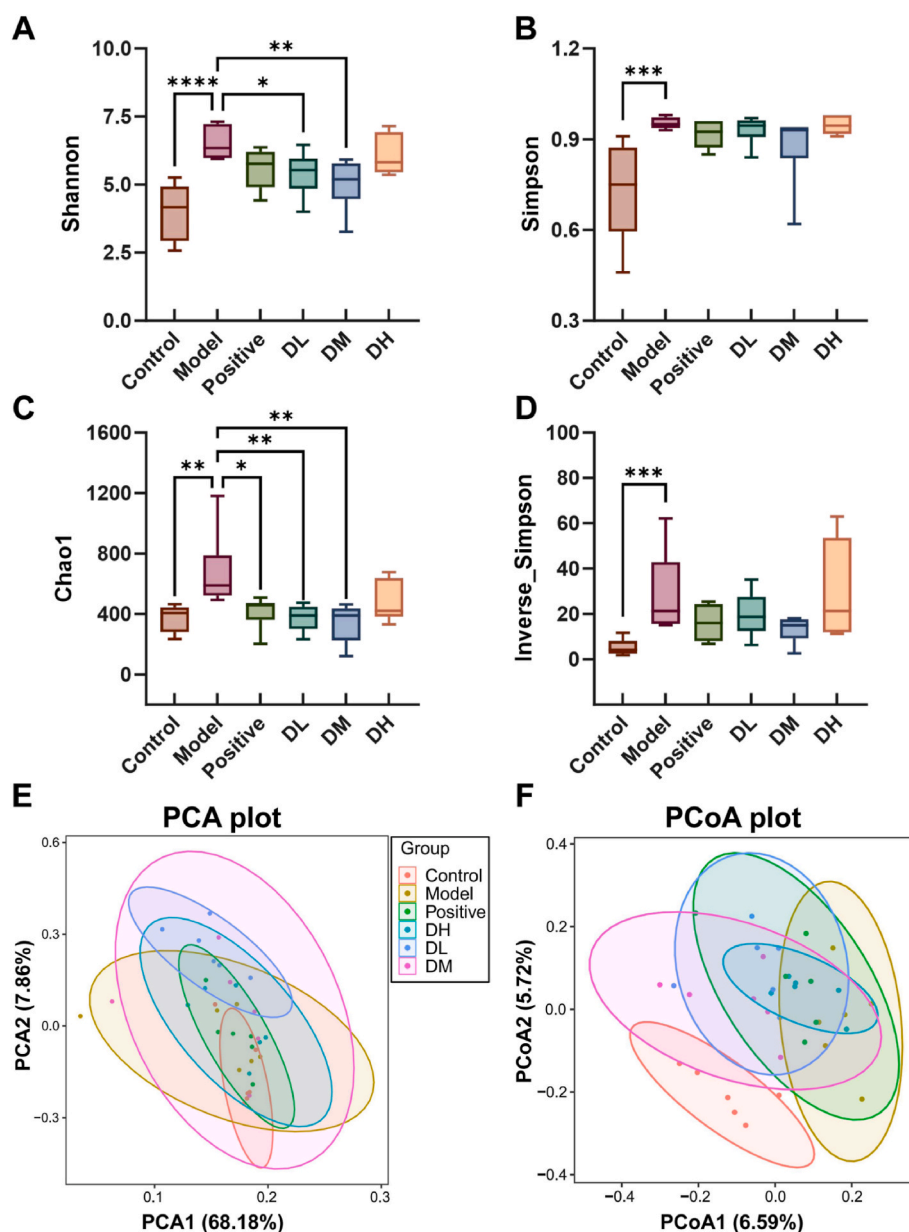


Fig. 8. Alpha and Beta Diversity of Gut Microbiota Among Different Groups of Mice. (A–D) α -diversity indices of the gut microbiota. Among these indices, (A) corresponds to the Shannon index, (B) represents the Simpson's index of diversity, (C) denotes the Chao1 index, and (D) signifies the Simpson's reciprocal index. Values within graphs are presented as median values with indication of the 25th and 75th percentiles and minimum and maximum values (represented as a boxplot), $n = 6$, and the significance difference was evaluated by Kruskal-Wallis test followed by uncorrected Dunn's test, each comparison stands alone. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. (E–F) β -diversity of the gut microbiota. (E) depicts the PCA plot, and (F) displays the PCoA plot. The abscissa represents one principal component, the ordinate represents another principal component, and the percentage values represent the proportion of the observed variation explained by the respective principal component. Each dot on the diagram represents a sample, and samples from the same group are in the same color. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

doses of ZSS-PR, a restorative trend was observed in the microbial community composition.

3.6.2. Analysis of gut microbiota composition

Figs. 9 and 10 present information on taxa composition at the phylum and genus levels, focusing on the top 30 and highlighting taxa with significant changes in relative abundance.

In Fig. 9, Firmicutes and Bacteroidota emerge as dominant phyla. Compared to the control group, the model group exhibits decreased relative abundance in Firmicutes, Verrucomicrobiota, Patescibacteria, Cyanobacteria, Fusobacteriota, and Acidobacteriota. Conversely, Bacteroidota, Actinobacteriota, Proteobacteria, Deferribacterota,

Campylobacterota, Planctomycetota, Desulfobacterota, and Chloroflexi show increased relative abundance. Following treatment with various doses of ZSS-PR, the relative abundance of all bacteria, except Actinobacteriota and Chloroflexi, demonstrates varying degrees of callback. Notably, Firmicutes, Bacteroidota, and Fusobacteriota exhibit the most significant changes ($p < 0.05$).

As shown in Fig. 10A, there were significant differences in the relative abundance of taxa among different groups, in which *Lactobacillus* and *Muribaculaceae_unclassified* stand out as the dominant taxa at the genus level. In order to reflect the changes of taxa between different groups as a whole, the similarities and differences of taxa between different groups were presented by heat map. On the longitudinal level,

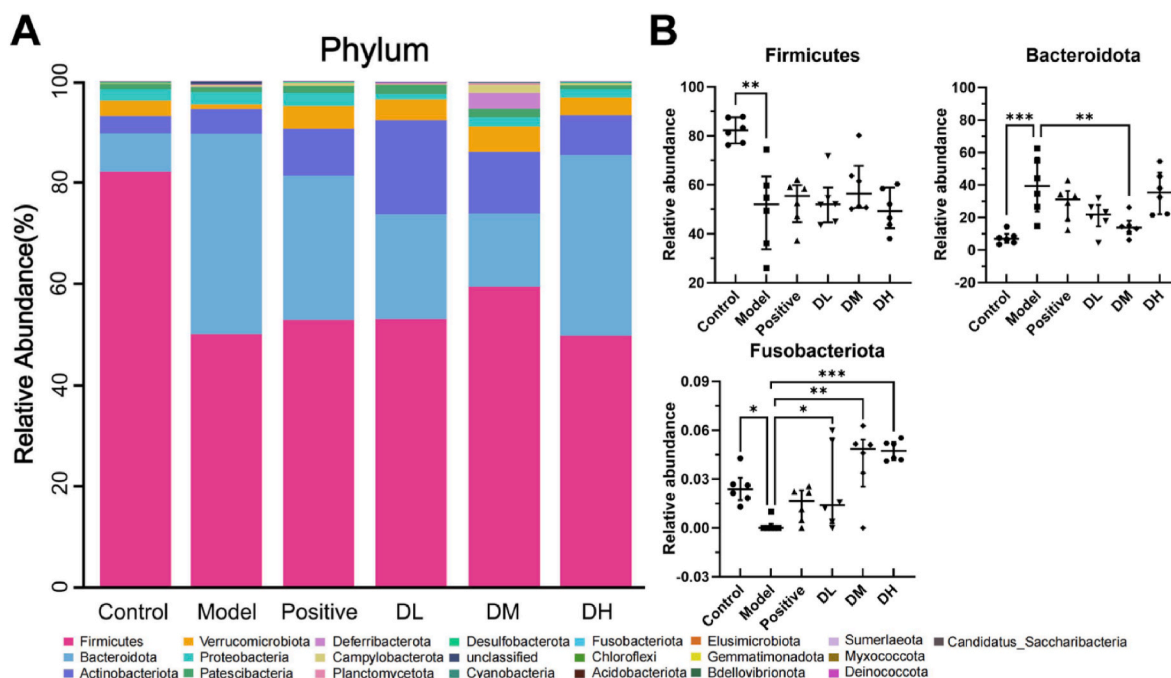


Fig. 9. Impact of ZSS-PR on Gut Microbiota Composition at the Phylum Taxonomic Level in Insomniac Mice. **(A)** Bar graph illustrating the relative abundance of species at the phylum taxonomic level. **(B)** Species significantly influenced by ZSS-PR at the phylum taxonomic level in insomniac mice. Values in the graphs are presented as median values with indications of the 25th and 75th percentiles, as well as minimum and maximum values (represented as a scatter plot), $n = 6$. Significance differences were assessed using the Kruskal-Wallis test, followed by uncorrected Dunn's test, with each comparison standing alone. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

as depicted in Fig. 10B, the control and ZSS-PR low and medium dosage groups exhibited similar clustering patterns, while the model and positive groups were clustered together. Compared to the model group, other groups displayed a tendency towards color variations that resembled those of the control group on a horizontal axis. This observation suggests that ZSS-PR is more effective than diazepam in regulating gut microbiota imbalance in insomnia animals. As shown in Fig. 10C, compared to the control group, the relative abundance of each genus in the model group demonstrates varying degrees of alteration. Notably, *Ligilactobacillus* and *Gardnerella* experience a significant reduction, while *Muribaculaceae_unclassified*, *Paramuribaculum*, and *Faecalibacterium* show a significant increase. Following the administration of different doses of ZSS-PR, all genera, except *Bifidobacterium*, *Lachnospiraceae_NK4A136_group*, and *Escherichia-Shigella*, exhibit varying correctional trends. Particularly, *Muribaculaceae_unclassified*, *Ligilactobacillus*, *Enterorhabdus*, *Paramuribaculum*, *Gardnerella*, and *Faecalibacterium* display the most prominent pullback trends ($p < 0.05$). These findings indicate that mice with insomnia exhibit an imbalance in their microbiota, and ZSS-PR can effectively rectify this imbalance.

3.6.3. Analysis of gut microbiota biomarkers

The Linear Discriminant Analysis Effect Size (LEfSe) was employed to assess the impact of sample abundance on observed differences, with only statistically significant marker taxa ($p < 0.05$, $\text{Lg (LDA Score)} > 4.0$) identified across different groups. The analysis results for Control vs Model and Model vs DM are presented. In Fig. 11A, for the Control vs Model comparison, LEfSe analysis revealed 12 hallmark taxa, including Firmicutes, Bacilli, Lactobacillales, Lactobacillaceae, *Lactobacillus*, and *Lactobacillus_sp_L_YJ* in the control group. Simultaneously, Bacteroidota, Bacteroidia, Bacteroidales, Muribaculaceae, and *Muribaculaceae_unclassified* were identified in the model group. Similarly, in Fig. 11B, for the DM vs Model comparison, LEfSe analysis identified 9 hallmark taxa. These included Bacteroidota, Bacteroidia, Gammaproteobacteria, Bacteroidales, Muribaculaceae, and *Muribaculaceae_unclassified* in the model group, while *Ligilactobacillus* and

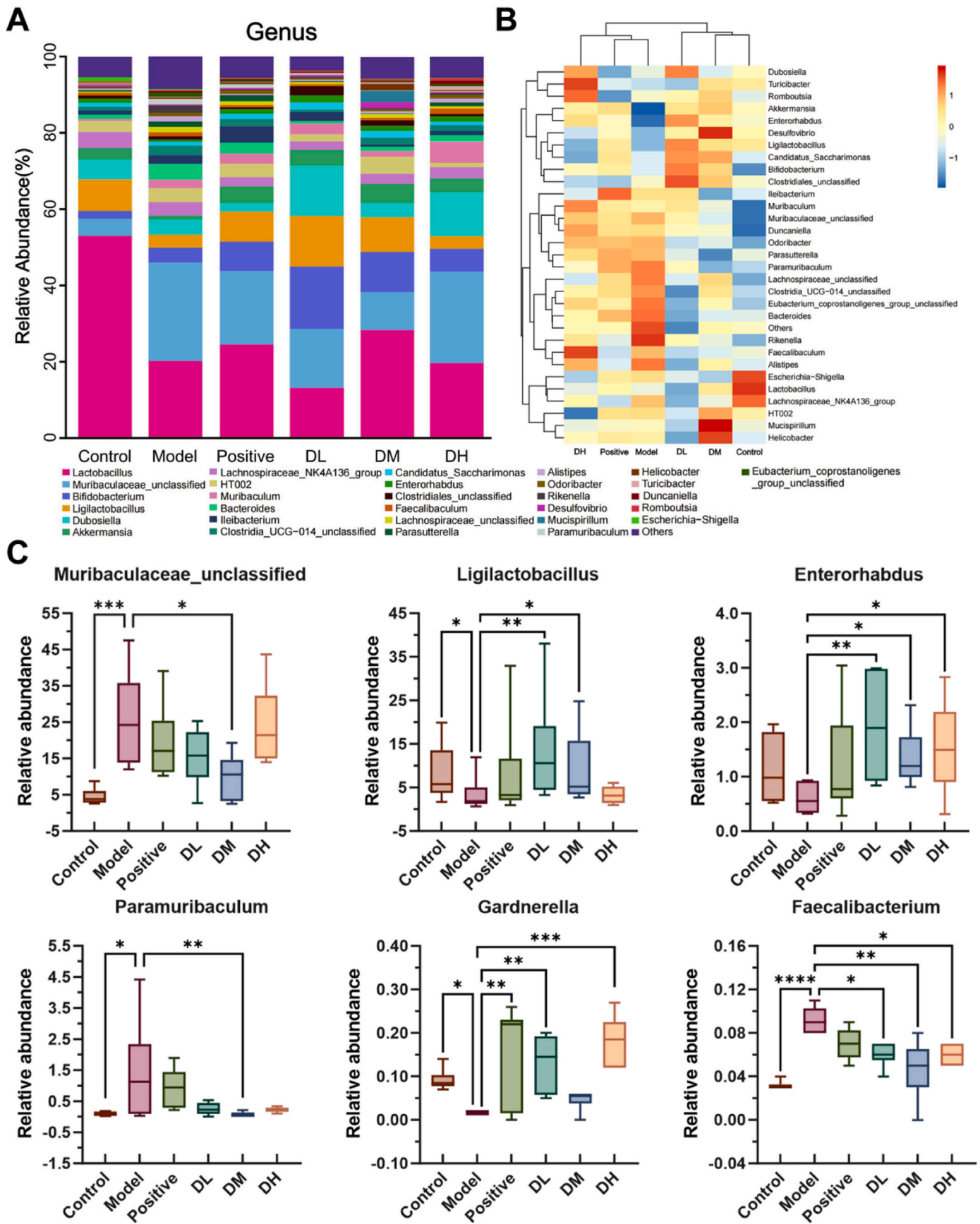
Ligilactobacillus_unclassified were found in the DM group.

Based on the aforementioned findings, the control group was distinguished by the presence of the marker taxa Firmicutes | Bacilli | Lactobacillales | Lactobacillaceae | *Lactobacillus* | *Lactobacillus_sp_L_YJ*. In contrast, the model group exhibited the marker taxa Bacteroidota | Bacteroidia | Bacteroidales | Muribaculaceae | *Muribaculaceae_unclassified* | *Muribaculaceae_unclassified*. Within the DM group, *Ligilactobacillus* | *Ligilactobacillus_unclassified* was identified as the marker taxa. The results revealed that ZSS-PR could increase the relative abundance of *Ligilactobacillus* in the microbiota of insomniac mice.

3.7. Analysis of the content of SCFAs

GC-MS technology was employed to analyze SCFA concentrations in cecal content samples from each animal group. Supplementary Fig. 5 depicts the Total Ion Chromatogram (TIC) graphs of the standards and Quality Control (QC) samples. Linear parameters for each SCFA can be found in Supplementary Table 3, and the validation results of the analytical methodology are presented in Supplementary Table 4.

The SCFA analysis results are illustrated in Fig. 12. Acetic, propionic, and butyric acids displayed the highest levels (0.4–3.7 $\mu\text{g}/\text{mg}$), while isobutyric, isovaleric, and valeric acids exhibited lower concentrations (0.01–0.12 $\mu\text{g}/\text{mg}$). Compared to the control group, the model group exhibited varying degrees of decrease in acetic, propionic, and butyric acid content, with isobutyric, isovaleric, and valeric acid content showing varying degrees of increase. Upon treatment with diazepam or different doses of ZSS-PR, the levels of the aforementioned SCFAs exhibited varying degrees of alteration. Diazepam significantly regulated isobutyric and valeric acid content. Administering different doses of ZSS-PR resulted in noticeable differences in SCFA levels. A low dose of ZSS-PR significantly reduced isobutyric, isovaleric, and valeric acid levels without significantly affecting other SCFAs. Conversely, medium and high doses of ZSS-PR significantly decreased isobutyric acid levels and increased acetic, propionic, and butyric acid levels in insomniac mice. The study results suggest a significant alteration in SCFA



(caption on next page)

Fig. 10. Impact of ZSS-PR on Gut Microbiota Composition at the Genus Taxonomic Level in Insomniac Mice. **(A)** Bar graph illustrating the relative abundance of species at the genus taxonomic level. **(B)** Intergroup cluster analysis of the relative abundances of taxa at the genus level. Each row represents the taxon, and each column represents the group. The gradient color from blue to red reflects the change of abundance from low to high, the closer to blue, the lower abundance, the closer to red, the higher abundance. The heatmap was transformed by Z value to normalize the expression abundance of the same taxon. **(C)** Relative abundance of species significantly affected by ZSS-PR at the genus taxonomic level. Values in the graphs are presented as median values with indications of the 25th and 75th percentiles, as well as minimum and maximum values (represented as a boxplot), $n = 6$. Significance differences were evaluated using the Kruskal-Wallis test, followed by uncorrected Dunn's test, with each comparison standing alone. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

production in mice with insomnia, effectively reversed through ZSS-PR treatment.

3.8. Integration analysis

Spearman correlation analysis was conducted to explore the relationship between gut microbiota at the phylum and genus taxonomic level (differential microbiota and marker microbiota that discriminated between model and treated group), biochemical markers (neurotransmitters, endocrine hormones, and cytokines), and metabolites. The results were visually presented in heat maps, as shown in Fig. 13A and B. In Fig. 13A, *Faecalibacterium*, *Muribaculaceae_unclassified*, *Bacteroidota*, and *Paramuribaculum* exhibit a strongly negative correlation with IL-1 β , TNF- α , MT, and 5-HT, while *Gardnerella*, *Firmicutes*, and *Lactobacillus* demonstrate a highly positive correlation with 5-HT and MT. Fig. 13B illustrates that *Faecalibacterium* and *Bacteroidota* show a highly positive correlation with hexanoylglycine but a strongly negative correlation with glutamic acid and hippuric acid. Furthermore, *Gardnerella* shows a highly negative correlation with 2-hydroxyglutaric acid and hexanoylglycine but a strongly positive correlation with glutamic acid. To investigate the correlation between gut microbiota and SCFAs, Redundancy Analysis (RDA) was performed, and the results are displayed in Fig. 13C and D. In Fig. 13C, *Firmicutes*, *Lactobacillus* and *Ligilactobacillus* are strongly correlated with acetic acid, propionic acid, and butyric acid, whereas *Muribaculaceae_unclassified*, *Paramuribaculum* and *Bacteroidota* show significant associations with isovaleric acid, valeric acid and isobutyric acid. The findings in Fig. 13D demonstrate that acetic acid, propionic acid, and butyric acid levels show substantial correlations with the gut microbiota in ZSS-PR-treated mice. Conversely, isovaleric acid, valeric acid, and isobutyric acid exhibit significant associations with the gut microbiota observed in the Model.

4. Discussion

In this study, ZSS-PR was prepared using the traditional decoction extraction method, which was in line with the clinical practice of Traditional Chinese Medicine. The chemical composition of ZSS-PR was determined, annotating a total of 152 components by comparing results with relevant literature. Some of these compounds are considered potentially active substances contributing to the sedative and hypnotic effects of ZSS-PR. For instance, jujuboside B supposed to exert a calming effect since it has been found to enhance the expression of GABAA receptors in cultured rat hippocampal neurons and to promote the opening frequency of chloride channels (Song et al., 2017). When combined with GABA, ZSS demonstrated superior in regulating metabolic patterns associated with sleep (Ren et al., 2023). Tenuifolin has been found to exhibit sleep-enhancing effects in mice by activating GABAergic systems and/or inhibiting noradrenergic systems (Cao et al., 2016). Prior research has identified flavonoids and saponins with high molecular weight as the primary constituents responsible for the sedative and hypnotic effects in ZSS-PR. Also in this study, a high number of flavonoid glycosides and triterpene saponins has been annotated in ZSS-PR. However, due to their limited ability to traverse the blood-brain barrier, alternative mechanisms, such as facilitating brain-gut interactions via gut microbiota, may be implicated in their sleep-regulating properties.

The secretion of neuroactive molecules and amino acid metabolism

in the neuroendocrine system are closely related to insomnia. Relevant neurotransmitters (e.g., 5-HT and DA) are primarily secreted by the gut enterochromaffin (EC) cells. These neurotransmitters regulated by the gastrointestinal tract are closely related to animals' emotional, cognitive and other mental activities of animals (Huang and Wu, 2021). Consistent with insomnia conditions, the contents of 5-HT and MT were significantly reduced in the duodenum of insomniac mice, accompanied by increased DA level (Fig. 4). After treatment with ZSS-PR, the above abnormal neurotransmitters and hormone levels were obviously brought back to normal levels. The 5-HT precursors tryptophan, DA precursors tyrosine and phenylalanine were also detected in the serum. However, there was no significant difference in the relative content of these precursors between the control and the model group (Fig. 6). The levels of tryptophan, tyrosine and phenylalanine are inconsistent with the corresponding neurotransmitters in each group, which may be related to disturbances in amino acid metabolism in insomniac animals. It has been reported that arginine synthase is beneficial for repairing pathological and functional disorders in rats exposed to REM sleep deprivation (Jiang et al., 2017). Consistent with this, in the present study, ZSS-PR could significantly normalize the low level of arginine in the serum of insomnia animals, and the mechanism of ZSS-PR improving insomnia through arginine needs to be further revealed. Glutamate, an essential excitatory neurotransmitter within the central nervous system, substantially influences sleep regulation. Perturbations in sleep patterns may arise when the secretion level of glutamate is either excessive or inadequate (Yu et al., 2019). In the present study, the abnormal glutamate level could be called back after treatment of insomnia mice with ZSS-PR. The sleep hormone MT is produced not only by the pineal gland but also by EC cells. The gut microbiota plays a protective role in cerebrovascular disease by activating the MT receptor, while the MT in turn, can also play a beneficial role in repairing gastrointestinal function or pathological abnormalities (Mehrzadi et al., 2023). The MT release is closely associated with circadian rhythm and sleep patterns (Zhang et al., 2017; Poza et al., 2022). Interestingly, circadian rhythmicity could also be observed in the intestinal microbiota (Kuang et al., 2019), and the efficacy and mechanism of ZSS-PR in treating insomnia through intestinal microbiota and MT circadian rhythm need to be further studied and revealed. Also in the present study, ZSS-PR was observed to increase the levels of beneficial biomarkers such as hippuric acid, while reducing the levels of harmful biomarkers such as uric acid, hexanoylglycine, 2-hydroxyglutaric acid, and pseudouridine. Hippuric acid is a microbial-host co-metabolite, and its elevated level is a marker of metabolic health. Also it was found to be positively correlated with gut microbiota diversity (Brial et al., 2021). Uric acid is a potentially critical factor in the pathogenesis of acute kidney injury, chronic kidney disease and hypertension. Some studies have also shown that short sleep duration is positively correlated with higher uric acid levels (Fathallah-Shaykh and Cramer, 2014; Chou et al., 2020). Hexanoylglycine is a biomarker of abnormal glycolipid metabolism (Boulangé et al., 2013). 2-Hydroxyglutaric acid is increased in specific inborn defects of metabolism and is now recognized as an oncogenic metabolite (Jones et al., 2017). Pseudouridine plays an essential role in the pathogenesis of diabetic nephropathy (Mathew et al., 2024). In the state of insomnia, in addition to perturbation of endocrine metabolism, the body can also show a decreased state of immunity, such as a decrease in the production of cytokines (Lu et al., 2022; Tang et al., 2019). Consistent with these literature reports, in the current study, the cytokine levels of TNF- α and

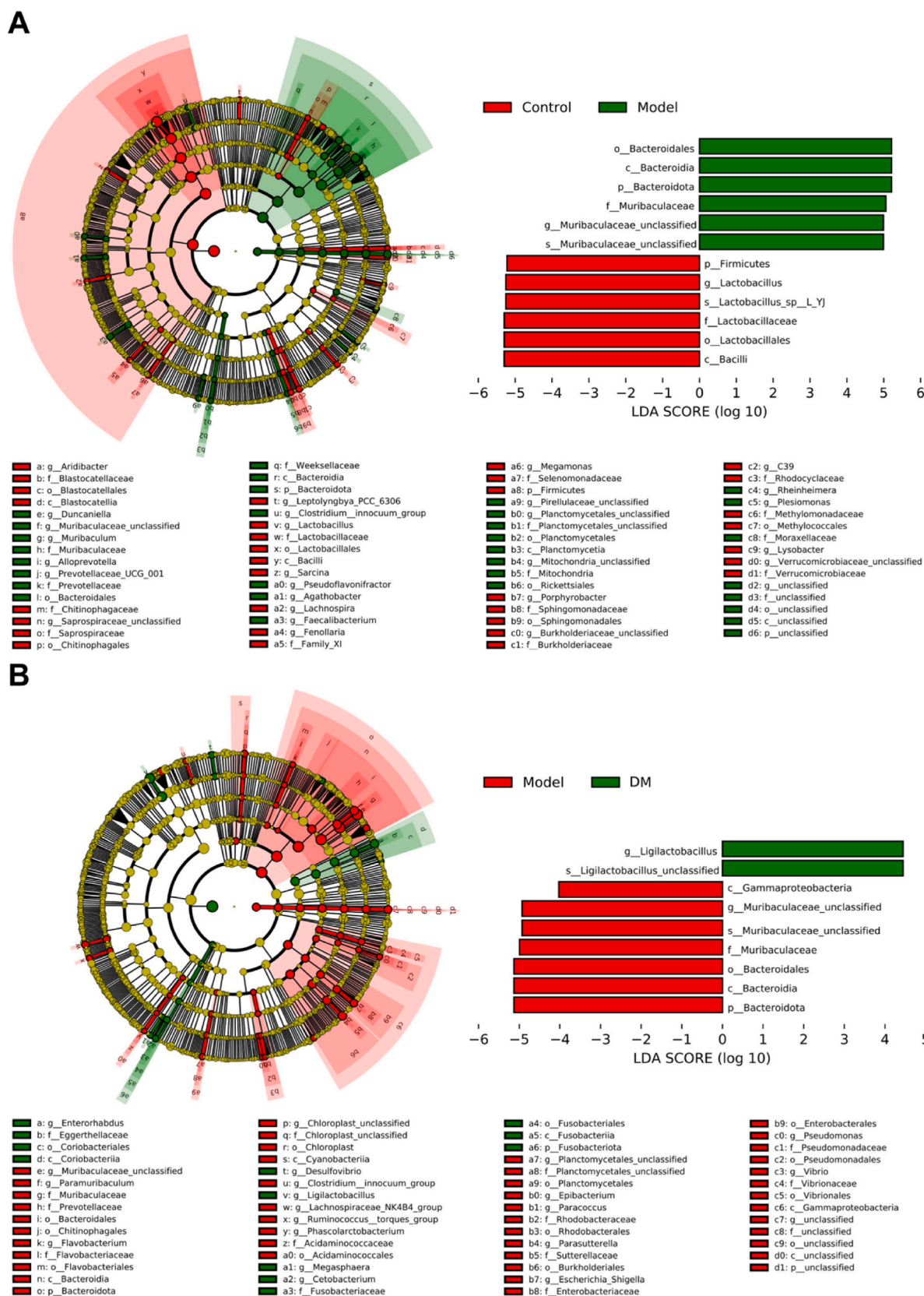


Fig. 11. Comparison of Intergroup Differences in Gut Microbiota Based on LDA and LefSe. **(A)** Cladogram and LDA scores generated from LefSe analysis for the Control vs Model (Kruskal-Wallis and Wilcoxon test, corrected by Benjamini Hochberg method, LDA >4, $p < 0.05$). **(B)** Cladogram and LDA scores generated from LefSe analysis for Model vs DM (Kruskal-Wallis and Wilcoxon test, corrected by Benjamini Hochberg method, LDA >4, $p < 0.05$). The cladogram's circles radiating from the inside out represent the classification level from phylum to species. The alphabetic species names are shown in the legend below. No significant difference is indicated in the species' uniform color yellow, and the biomarker follows differences between species groups in staining. The length of the bar chart represents the impact of different species. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

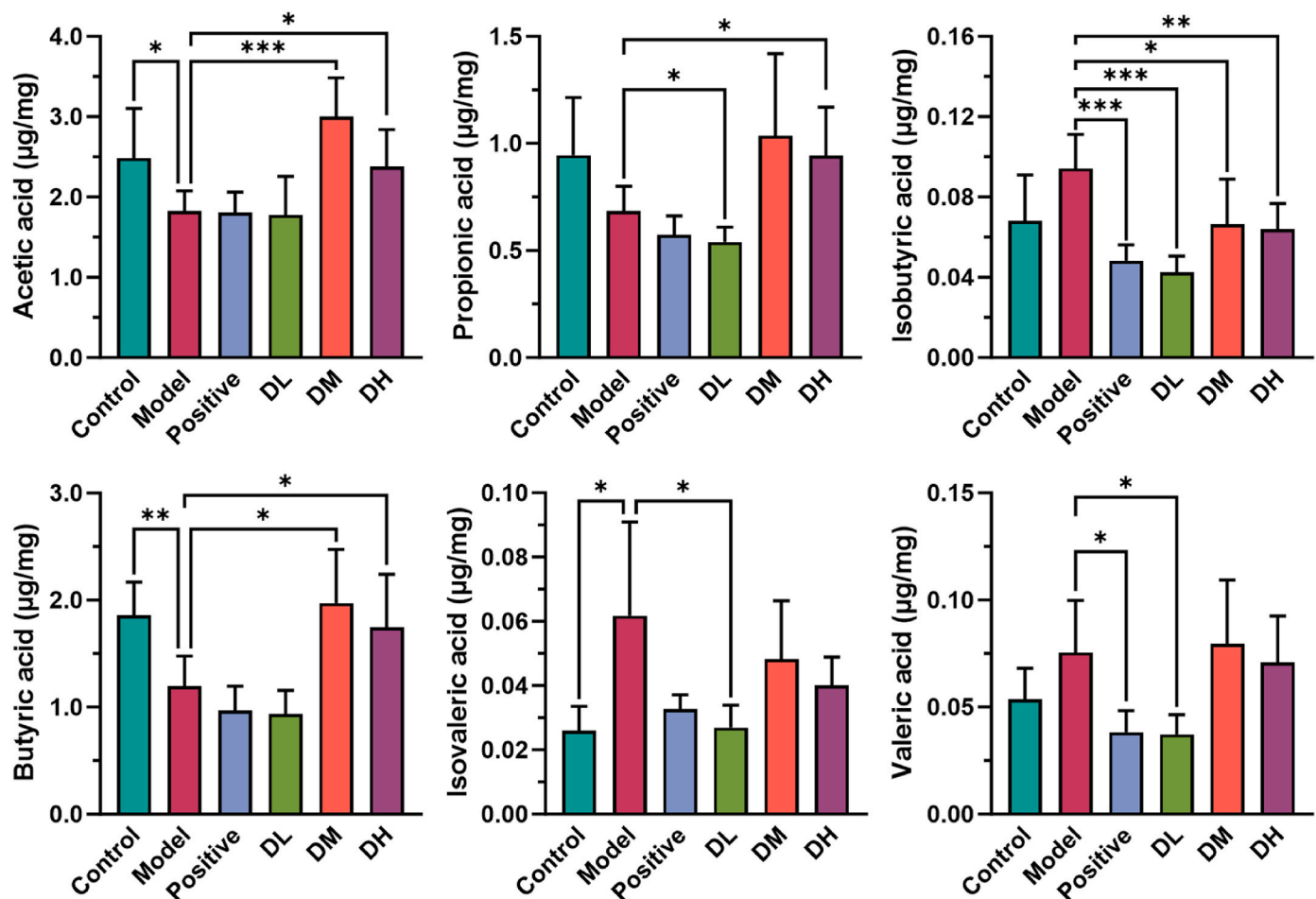


Fig. 12. Effect of ZSS-PR on SCFA Levels in Cecal Contents of Insomniac Mice. Values are presented as the mean $\pm$ SD, $n = 6$. Normal distribution was verified by Shapiro-Wilk test. Brown-Forsythe and Welch ANOVA, followed by Dunnett's T3 multiple-comparison test, were performed for statistical analysis. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

IL-1 β were reduced in the serum of insomnia animals, while ZSS-PR could significantly restore cytokine levels. The findings of other scholars showed that similar to the regulation of melatonin, the levels of these inflammatory cytokines in insomnia are also regulated by circadian rhythm (Vgontzas et al., 2002), which, to some extent, explains the differences in the increase or decrease of inflammatory factors reported by different researchers in the state of insomnia.

Gut microbiota can also influence sleep via the production of SCFAs (Wang et al., 2022). Firmicutes and Bacteroidota are the two phyla that account for the most significant proportion of the human gut microbiome. Studies have shown notable changes in the gut microbiota composition in individuals with insomnia, especially regarding Firmicutes and Bacteroidota (Hua et al., 2021). The results of other scholars showed that the relative abundance of Firmicutes increased while Bacteroidota decreased under the condition of insomnia (Liu et al., 2019). The gut microbiota is a complex and delicate microbial regulatory system in the body's intestinal tract (Yang et al., 2023), consistent with the TCM theory of Yin and Yang balance. Under normal circumstances, the colonised gut microbiota maintain a steady-state balance that restricts each other (Li et al., 2023). Therefore, it is speculated that dysbalance in microbiome composition can also lead to gastrointestinal dysfunction and mental abnormalities through the neuroendocrine-immune network (Tang and Cao, 2021). The current findings support this perspective, revealing a significant increase in the relative abundance of Bacteroidota, along with significant decreases in Firmicutes and Fusobacteriota in the insomniac mice, consistent with previous reports (Du et al., 2022; Liu, 2021). Additionally, significant modifications were observed

at the genus level, specifically in the relative abundance of *Muribaculaceae_unclassified*, *Ligilactobacillus*, *Gardnerella*, and *Faecalibacterium* in mice with insomnia. According to the LEfSe analysis in Fig. 11, *Ligilactobacillus* was a potential biomarker for ZSS-PR treatment of insomnia. There is increasing evidence that *Ligilactobacillus* is widely involved in host intestinal metabolism and immune activities, and its reduction is highly correlated with ageing (Zhou et al., 2024; González-Muniesa et al., 2024). As metabolites produced by the microbiota, SCFAs play a vital role in mediating the brain-gut axis and are closely associated with neurotransmitter levels regulating the host's sleep patterns (Chen et al., 2020; Singh et al., 2017). Extensive research consistently demonstrates a strong association between SCFAs and the host's nervous and immune systems (Ikeda et al., 2022). Acetic acid and butyric acid can potentially enhance sleep quality through diverse mechanisms (Szentirmai et al., 2019). In contrast, propionic acid can modulate neurotransmitter levels, including Glu, GABA, and tryptophan, impacting emotional regulation, cognitive processes, and behavioral responses (DeCastro et al., 2005). Butyrate has been observed to exert a favorable regulatory influence on sleep modulation and maintaining intestinal homeostasis (Leonel and Alvarez-Leite, 2012; Szentirmai et al., 2019). Aligned with an existing report (Wang et al., 2021), the present study demonstrated a notable reduction in acetic acid and butyric acid contents alongside an elevated isovaleric acid level in mice with insomnia. It is hypothesized that the alteration in SCFA content may be attributed to the dysregulation of intestinal microbiota. Administering ZSS-PR treatment effectively reinstated the aforementioned SCFA levels to their baseline values. This finding suggests that the

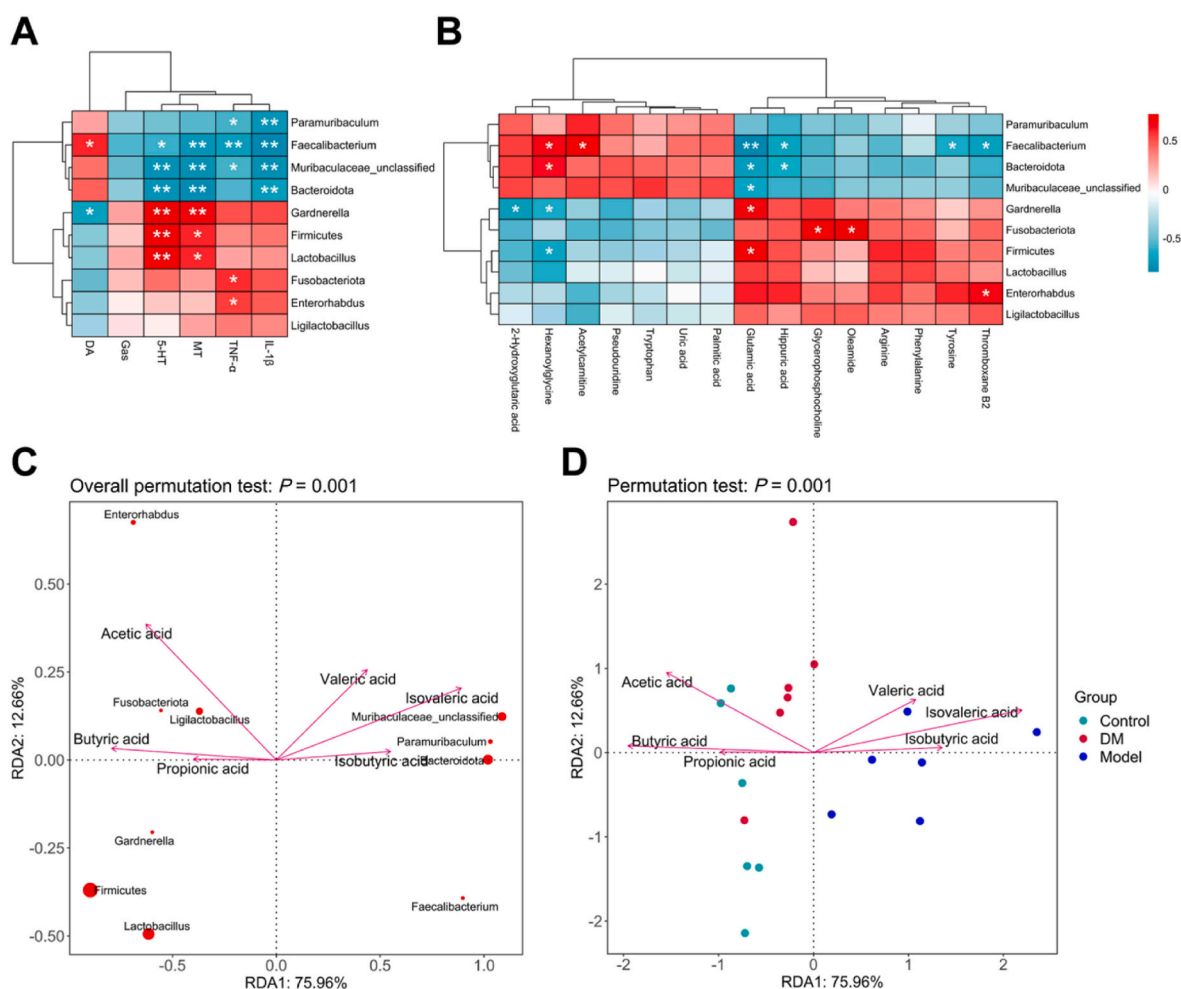


Fig. 13. Correlation Analysis Between Gut Microbiota, Biochemical Markers, Metabolites, and SCFAs. **(A)** Correlation analysis between levels of biochemical markers (5-HT, DA, Gas, MT, TNF- α , and IL-1 β) and gut microbiota at the phylum and genus taxonomic level. **(B)** Correlation analysis between the levels of metabolites and gut microbiota. Spearman correlation was conducted to calculate statistical significance, and FDR correction was performed using Benjamini and Hochberg. Statistical significance between groups is indicated as * $p < 0.05$, ** $p < 0.01$. Results are visually presented, with red indicating a positive correlation and blue indicating a negative correlation. The intensity of colors corresponds to the correlation coefficient's magnitude, where darker shades represent stronger correlations and lighter shades represent weaker ones. **(C)** Correlation analysis between the levels of SCFAs and gut microbiota. **(D)** Correlation analysis between SCFAs and various experimental groups. Redundancy analysis (RDA) was conducted to calculate the correlations, and Permutational multivariate analysis of variance (PERMANOVA) was used to assess the statistical significance. The scatter's magnitude indicates the correlation strength, with a greater scatter indicating a more robust correlation. Arrows represent SCFAs, and their placement within a quadrant indicates the correlation between SCFAs and the sorting axis. The length of the line connecting each arrow to the origin indicates the magnitude of the correlation between specific SCFAs and gut microbiota, with a longer line signifying a more robust correlation. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

restoration of gut microbiota dysbiosis led to a restoration in the production of SCFAs, implying that ZSS-PR can modulate sleep patterns through its influence on microbiota and SCFA production. Nonetheless, the intervention effect of varying doses of ZSS-PR did not exhibit a dose-dependent relationship. We speculate that this may be related to the complex components and multi-target regulatory effects of ZSS-PR (Zhao et al., 2023), which also needs to be investigated.

In summary, this study employed PCPA-induced insomniac mice as the experimental model to investigate the correlation between insomnia and endogenous metabolism and the balance of gut microbiota. We explored the underlying mechanism of ZSS-PR's efficacy in treating insomnia by integrating metabolomics and gut microbiota analysis. According to the findings depicted in Fig. 14, the administration of ZSS-PR demonstrated the ability to regulate sleep patterns in mice with insomnia. This effect was achieved by addressing abnormal autonomic activity observed in insomniac mice, reducing pathological damage in associated brain regions, modulating abnormal endogenous metabolic profiles, ameliorating imbalances in gut microbiota, and counteracting

the aberrant production of SCFAs. Based on the results, the correlation analysis between the above indices showed that the gut microbiota regulation by ZSS-PR in insomnia was related to restoring normal levels of enteric neurotransmitters, endocrine hormones, cytokines, endogenous metabolites and SCFAs. The results of this study preliminarily outline the brain-gut regulation pathway of ZSS-PR in treating insomnia, and its related molecular mechanism needs to be further revealed. As indicated by the previous literature, the neuroendocrine-immune factors, i.e. MT, 5-HT, cytokines, and gut microbiota, all demonstrated circadian rhythm regulation activity. ZSS is a commonly used traditional Chinese medicine in treating insomnia with heart-liver and blood deficiency. In the clinic, ZSS shows a better sleep-promoting effect when the body is at rest while not producing a sleepy effect similar to sedative hypnotic drugs such as phenobarbital. However, the regulatory effect and mechanism of ZSS-PR pair on the circadian rhythm of these factors need to be further studied and revealed.

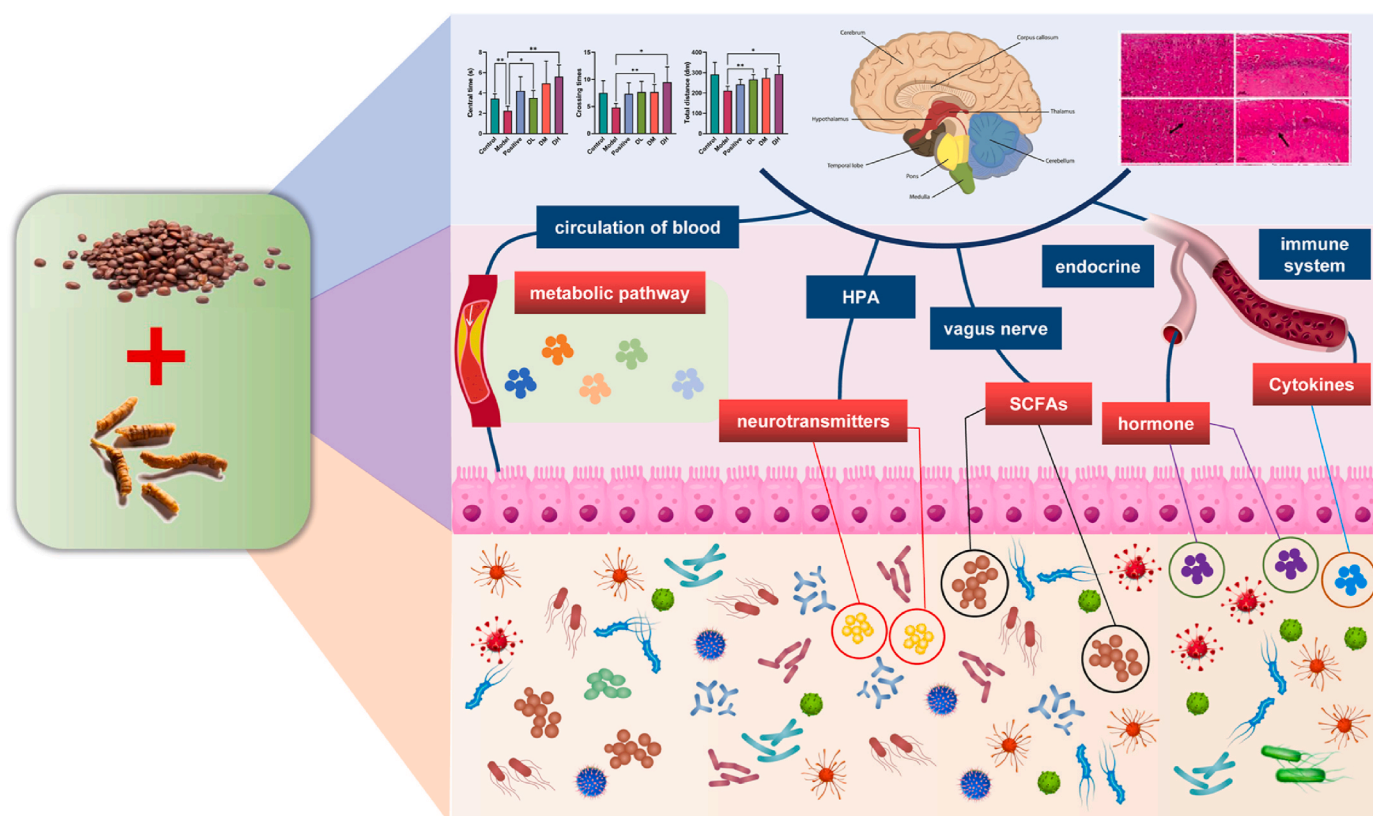


Fig. 14. Schematic Overview of the intervention mechanism of ZSS-PR for mice afflicted with insomnia.

5. Conclusions

The regulatory mechanism of ZSS-PR was investigated in an animal model of insomnia using metabolomics and gut microbiota analysis. Following PCPA administration, mice with insomnia exhibited notable changes in spontaneous activity, biochemical markers (including neurotransmitters, endocrine hormones, and cytokines), serum metabolic profiles, gut microbiota composition, and SCFA levels. The administration of ZSS-PR significantly improved sleep activity in mice with insomnia. This improvement involved restoring abnormal spontaneous activity, repairing pathological brain damage associated with insomnia, reversing gut microbiota imbalance, correcting abnormal SCFA levels, restoring abnormalities in the serum metabolome, and modulating abnormal levels of neurotransmitters, endocrine hormones, and cytokines in the gut. Moreover, the observed correlation between gut microbiota and various physiological factors such as neurotransmitters, endocrine hormones, cytokines, metabolites, and SCFAs suggests the brain-gut axis's involvement in ZSS-PR's sleep activity regulation in mice with insomnia. Further research is necessary to comprehensively understand the precise regulatory mechanisms underlying the effects of ZSS-PR.

CRediT authorship contribution statement

Haiqin Ren: Writing – original draft, Investigation, Conceptualization. **Xiangpeng Kong:** Writing – review & editing, Methodology, Investigation. **Yue Zhang:** Resources, Investigation. **Fanying Deng:** Investigation, Formal analysis. **Jianli Li:** Investigation, Data curation. **Fuxia Zhao:** Visualization, Investigation. **Pei Li:** Investigation, Data curation. **Ke Pei:** Resources, Investigation. **Jinyan Tan:** Investigation, Formal analysis. **Yangang Cheng:** Resources, Investigation. **Yan Wang:** Resources, Methodology. **Lu Zhang:** Supervision, Investigation. **Yingli Wang:** Writing – review & editing, Project administration,

Conceptualization. **Xuliang Hao:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Abbreviation

TCM	Traditional Chinese Medicine
ZSS	Ziziphi Spinosae Semen
PR	Polygalae Radix
ZSS-PR	Ziziphi Spinosae Semen and Polygalae Radix
5-HT	5-hydroxytryptamine
DA	dopamine
Gas	gastrin
MT	melatonin

TNF- α	tumor necrosis factor- α
IL-1 β	interleukin-1 β
SCFAs	short-chain fatty acids
GC-MS	Gas chromatography-mass spectrometry
Control	control group
Model	insomnia model group
Positive	positive group
DL	low-dose ZSS-PR treatment group
DM	medium-dose ZSS-PR treatment group
DH	high-dose ZSS-PR treatment group
HED	the human equivalent dose
PCPA	DL-4-Chlorophenylalanine suspension
OFT	Open-Field Test
H&E	Hematoxylin and Eosin
PCA	principal component analysis
OPLS-DA	orthogonal partial least squares-discriminant analysis
VIP	variable importance plot
MTBE	methyl tert-butyl ether
ASV	amplicon sequence variants
LEFSe	the Linear Discriminative Analysis Effect Size
RDA	Redundancy Analysis

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2024.118255>.

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